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Reversed Polarity Characteristic Magnetizations in the Albian Contessa Section, Umbrian Apennines, Italy: Implications for the Existence of a Mid-Cretaceous Mixed Polarity Interval

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Paleomagnetic and paleontologic data from the Valle della Contessa (Umbrian Apennines, Italy) span a 14 m.y. gap in previous magnetostratigraphic sections and reveal several intervals of reversed characteristic remanent magnetization (ChRM) in limestones and marls of the Albian Marne a Fucoidi. Rock magnetic data indicate that the reversed ChRM is carried by hematite having high unblocking temperatures (~625-690°C). Hematite with these unblocking temperatures also carries a normal polarity ChRM in other intervals. A potential bias towards the identification of normal polarity is present since at every horizon measured a normal polarity component, carried by magnetite and hematite, is isolated at lower unblocking temperatures (~300-600°C). Bulk magnetic properties vary throughout the measured section. In the portion containing most of the reversed ChRM intervals, hematite is the dominant carrier of natural remanent magnetization, while magnetite is the dominant carrier stratigraphically above and below. These changes correspond approximately with lithology. The zone of dominant hematite magnetization is marked by reddish bands (pigmentary hematite) which can be correlated thoughout Umbria and are thought to record changes in the oxidation state of the seafloor during deposition. Two models may account for the observed magnetization directions. The reversed magnetizations may record a remagnetization which occurred at least 18 m.y. after deposition, while the beds were still flat-lying, in reversed polarity chron 33R (83-79 Ma) or later. If the reversed ChRM represents a remagnetization, the process is potentially of great importance since it can produce magnetization patterns which resemble polarity intervals. Alternatively, the reversed magnetizations may have been acquired during intense seafloor oxidation episodes during the mid-Albian (107-104 Ma) and may record unrecognized intervals of reversed geomagnetic field polarity. Milankovitch-like bedding cyclicity can be used to tune the sedimentation rate and obtain estimates of the duration of the potential reversed intervals. Using these estimates, two of the potential reversed polarity intervals are of sufficient duration (>100 kyr) to be recognizable in both detailed stratigraphic sections and marine magnetic anomaly surveys. A primary magnetization model predicts that several intervals of reversed magnetization should be found near the boundary of the *Biticinella breggiensis* and *Ticinella primula* foraminiferal zones and within the *Prediscosphaera cretacea* nannofossil zone on a worldwide basis.

INTRODUCTION

Our knowledge of the mid-Cretaceous geomagnetic reversal chronology, originally derived from the interpretation of marine magnetic anomalies [*Larson and Pitman*, 1972; *Larson and Hilde*, 1975], has been confirmed by magnetostratigraphic study of uplifted Tethyan marine sequences [*Lowrie et al.*, 1980*a,b*; *Lowrie and Alvarez*, 1977]. These studies have defined two reversed polarity zones, corresponding to chrons 33R and M0, bounding an ~34-m.y.-long interval of normal polarity, the Cretaceous Normal Polarity Superchron (K-N). The lack of reversals within the K-N Superchron limits our attempts to correlate the striking geological characteristics of these mid-Cretaceous sections, such as periods of oceanic anoxia, which have been of broad interest to the geologic community.

One of the most thorough magnetostratigraphies defining the mid-Cretaceous time scale is that derived from the Gubbio-Cismon composite section (Figure 1) [*Lowrie et al.*, 1980*b*]. Yet this composite section has its limitations in coverage as a gap of approximately 14 m.y. interrupts the Cismon section [*Channell et al.*, 1979*a*] removing upper Aptian-upper Albian sediments (Figure 2). Two magnetostratigraphic sections from the Umbrian Apennines and one from the Southern Alps partially fill this gap. When paleomagnetic sampling has been dense (i.e., 1 specimen/50 cm or greater), brief reversed polarity intervals have sometimes been found, reversed intervals which are not included in most geomagnetic time scale syntheses [e.g., *Harland et al.*, 1982; *Kent and Gradstein*, 1985, *Harland et al.*, 1990].

Paleomagnetic sampling of the Valdorbia section [*VandenBerg et al.*, 1978; *Lowrie et al.*, 1980*b*] of the Umbrian Apennines covered ~2 m.y. of the lower part of the gap (upper Aptian) and defined one brief reversed polarity interval. Recently this zone, the ISEA reversed polarity interval, has been shown to correlate on a worldwide basis [*Tarduno et al.*, 1989; *Tarduno*, 1990]. Only one magnetostratigraphic section in the Apennines covers the majority of the mid-Cretaceous sampling gap (that of middle-late Albian age), the Poggio le Guaine section [*Lowrie et al.*, 1980*a*]. Sampling, however, was extremely sparse (seven samples in 25 m) since the emphasis in that study was on Early Cretaceous magnetostratigraphy. Paleomagnetic sampling of the Cerro Veronese section from the Southern Alps (northern Italy) [*Channell et al.*, 1979*a*] covered a 7-m interval of the gap with 14 hand samples, but middle Albian-upper Albian sediments were not sampled. Two reversed polarity intervals were found in the Valle del Mis section of the Southern Alps [*VandenBerg and*

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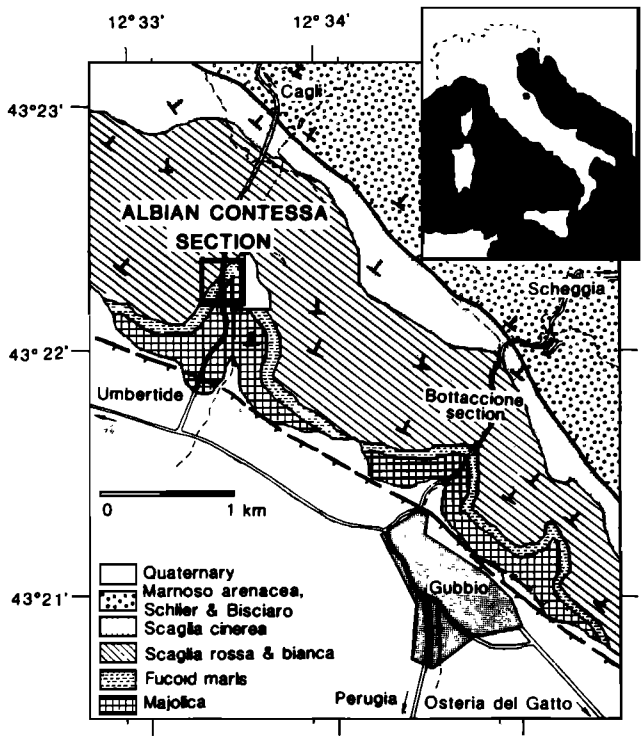


Fig. 1. Location map, Albian Contessa section, Umbrian Apennines, Italy.

Wonders, 1980] from siliceous limestones thought to be of middle-late Albian age. Recent reexamination of this section, however, has suggested that most of these limestones may be much older (E. Erba, personal communication, 1990).

In addition to the lack of a thorough magnetostratigraphic section in mid-Albian Tethyan sediments, other data sets indicate that our picture of mid-Cretaceous geomagnetic polarity may be incomplete. Paleomagnetic data from sediment cores recovered by Deep Sea Drilling Project (DSDP) leg 26 in the Indian Ocean were interpreted by *Green and Brecher* [1974] and *Jarrard* [1974] as evidence for several brief intervals of reversed polarity in the Albian. Based on this interpretation, *van Hinte* [1976] defined a "site 263" mixed polarity interval in the mid-Cretaceous. *Hailwood et al.* [1980] interpreted paleomagnetic data from the Bay of Biscay (DSDP leg 40) as representing several "short-period reversal events" in the Albian and correlated these with the "site 263" mixed polarity interval. More recently, *Urrutia-Fucugachi* [1988] has interpreted data from a preliminary magnetostratigraphic study of the Morelos Formation (southern Mexico) as recording a reversed polarity "event" in the Late Albian.

To fill the sampling gap in the Tethyan sections and to evaluate the existence of a mid-Cretaceous mixed polarity interval, new paleomagnetic and paleontologic specimens were collected from the Valle della Contessa of the Umbrian Apennines (Figure 1). This locality is well known for its lithostratigraphy [e.g., *Coccioni et al.*, 1989], cyclic sedimentation [*Schwarzacher and Fischer*, 1982] and Paleogene magnetostratigraphy [*Lowrie et al.*, 1982].

LITHOSTRATIGRAPHY AND PALEOMAGNETIC SAMPLING

Deposition of the Cretaceous stratigraphic sequence of the Umbrian Apennines [*Bortolotti et al.*, 1970] is thought to have occurred along the southern margin of the Tethys on a promontory of the African Plate known as Adria [*Channell et al.*, 1979*b*]. Subsequent closure of the Tethys and deformation related to the formation of the Apennines have exposed a sequence of pelagic rocks having ideal properties for magnetostratigraphic analysis [*Lowrie and Heller*, 1982]. A magnetostratigraphic type section for the Upper Cretaceous-Paleocene has been established in this sequence in the Bottaccione gorge near Gubbio [*Alvarez et al.*, 1977]. In Umbria, the middle to late Albian is represented by the middle to upper portion of the Marne a Fucoidi (also known as the Scisti a Fucoidi or Fucoid Marls), a series of marls and marly limestones punctuated by occasional black shales. The marls take their name from "Fucoidi", the local name of the ichnofossil *Chondrites* which is abundant in the formation [*de Boer and Wonders*, 1984; *Tornaghi et al.*, 1989]. The marls grade upwards into the upper Albian Scaglia Bianca, a white to grey limestone.

The Marne a Fucoidi, which rests on lower Aptian Maiolica Limestone, has been divided into the following six members [*Coccioni et al.*, 1987, 1989, 1990]: (1) the Greenish-Grey Cherty Member; (2) the Lower Reddish Member; (3) the Brownish Clayey Member; (4) the Greenish Marly Member; (5) the Upper Reddish Marly Member; and (6) the Whitish Marly Limestone Member. The spectacular changes in sediment color displayed by the Marne a Fucoidi have been interpreted as an indicator of the oxidation state of the seafloor during deposition [*Arthur and Fischer*, 1977; *Tornaghi et al.*, 1989]. Locally, detachment thrusts have been developed in the Marne a Fucoidi [*Alvarez et al.*, 1978]. Together with weathering, these faults render many outcrops unsuitable for magnetostratigraphic analysis.

In July 1989, 52 paleomagnetic cores and 20 hand samples were collected in a section through the Marne a Fucoidi in the Valle della Contessa, approximately 2 km northwest of the Bottaccione gorge [*Tarduno*, 1989] (Figure 1). The section was measured in the southern cement quarry where an unusually fresh face was exposed by the quarrying operation. The face sampled is apparently the same illustrated by *Coccioni et al.* [1989, pp. 572-573, illustration D] in their study of the Selli Level, a prominent early Aptian black shale interval which has recently been shown to correlate with similar deposits in the Pacific [*Sliter*, 1989; *Tarduno et al.*, 1989]. In the Contessa Quarry, the Selli interval is involved in a zone of local structural complexity which may have formed by back thrusting or slumping [*Coccioni et al.*, 1989]. Our section was measured from the base of two prominent thicker bedded marls (labeled 20 m) stratigraphically above this disturbed interval. The section extends through an interval of reddish marls which can be further subdivided into two reddish bands approximately 1-1.5 m thick, separated by 4-5 m of greenish marls. The rocks generally become more indurated upsection. The section ends in the transition from the Marne a Fucoidi to the Scaglia Bianca Limestone. The interval sampled corresponds to members 4 to 6 of *Coccioni et al.* [1990].

Many of the marls were difficult or impossible to collect by drilling. Therefore, our collection preferentially sampled the most indurated marl or limestone beds. The section was recollected in October 1989 to improve the sampling density. In addition, samples were collected in the same stratigraphic horizon to check the lateral persistence of directions. For much of the interval of interest, the stratigraphic sampling interval is roughly 1 sample per 30 cm.

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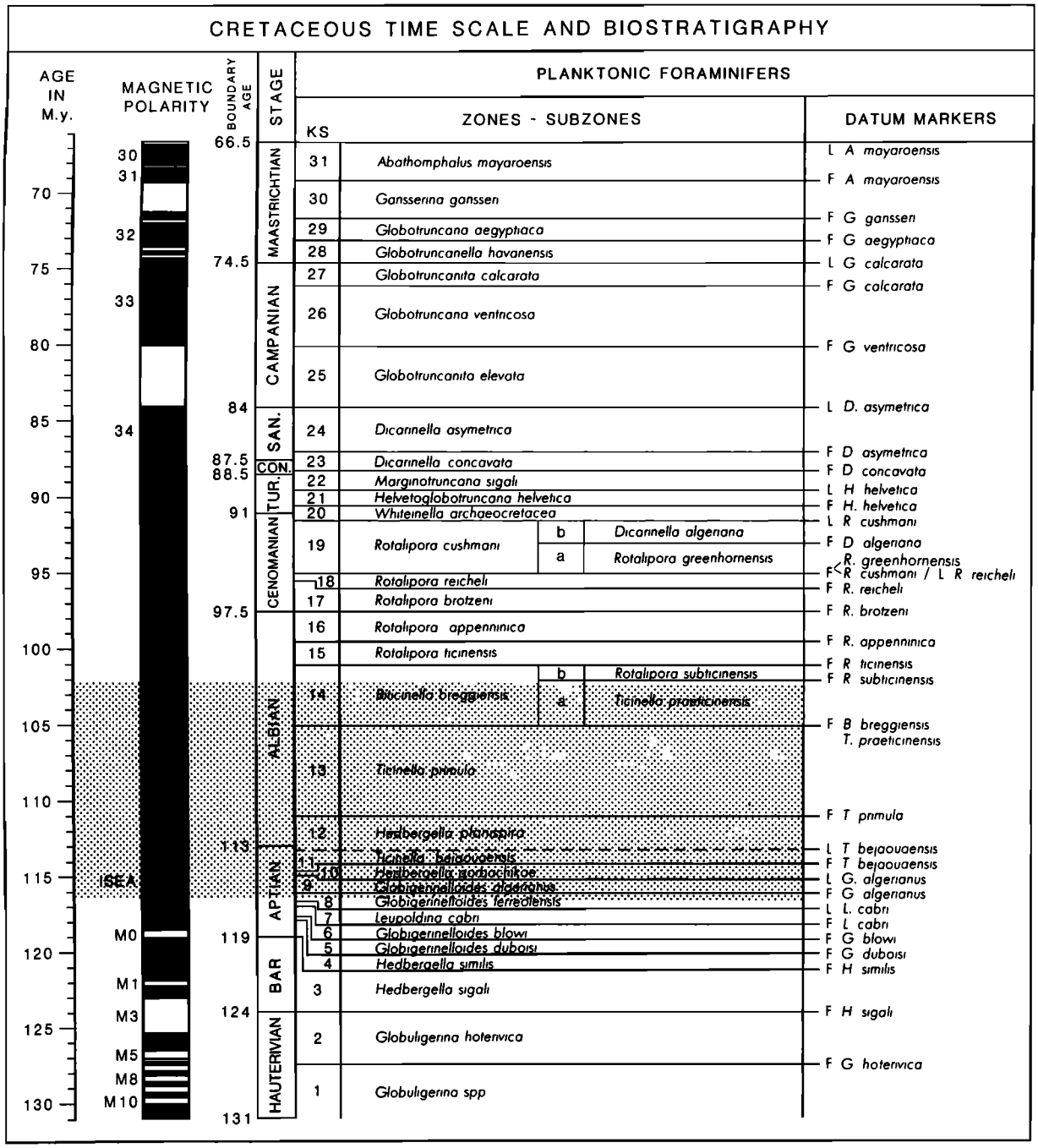


Fig. 2. Sampling gap in mid-Cretaceous magnetostratigraphy. Foraminiferal biostratigraphy after *Sliter* [1991]. F, first occurrence; L, last occurrence. The "hiatus" in the Cismon section [*Lowrie et al.*, 1980*b*] is approximately 14 m.y. (stippled), suggesting that a rigorous evaluation of geomagnetic polarity in Italian sections is not available for nearly 1/3 of the previously defined K-N Superchron.

Two features cut the section, and although possible gaps and/or overlaps are <1 m, this scale will be important in the following discussions. Feature B (Figure 3) is a high-angle fault. Feature A is man-made: a ramp in the quarry face. In addition to these features, it is possible that there are other small-scale faults, having several decimeters of offset, that were not recognized during sampling. Hand samples were taken stratigraphically above the ramp in the quarry face due to the danger of rockfalls presented by the uncleaned nature of the face. All samples stratigraphically below the ramp were collected as field-drilled cores. Subsamples were taken from the paleomagnetic samples for analyses of foraminifera and nannofossils.

MICROPALEONTOLOGICAL DATING

Planktic Foraminifers

Planktic foraminifers were analyzed in 20 thin sections cut from paleomagnetic samples (20.14 m to 50.05 m). The identified foraminifers are placed in three Albian planktic foraminiferal zones (Figure 4) according to the zonation of *Sliter* [1991]. The section ranges from the *Ticinella primula*

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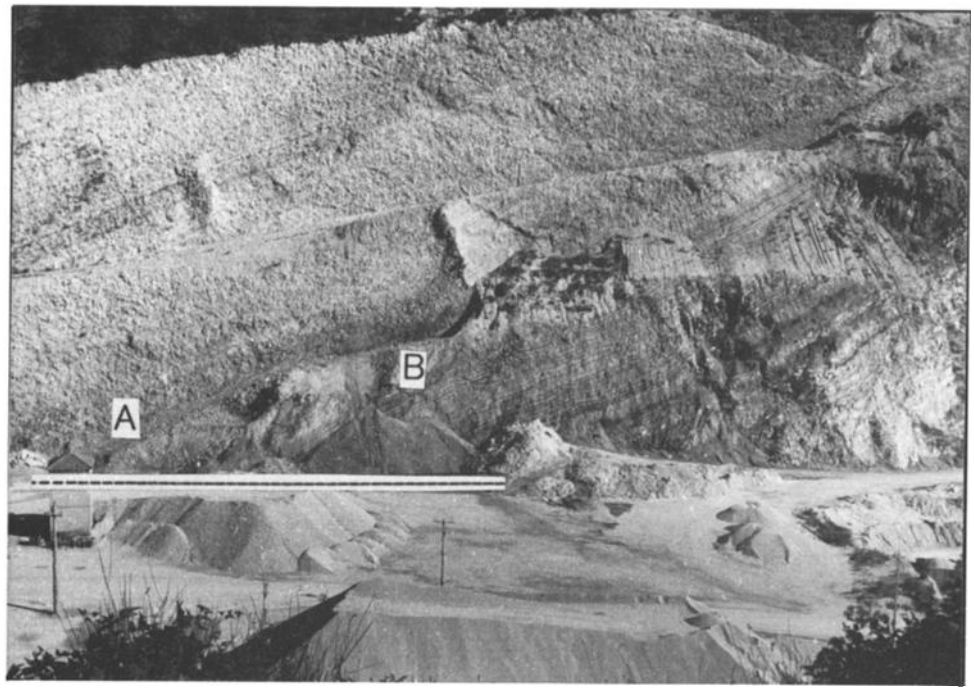


Fig 3. Albian sedimentary rocks in the Valle della Contessa, Umbrian Apennines, Italy. The sampled section (dashed) begins in the Marne a Fucoidi and ends in the transition zone to the Scaglia Bianca Limestone. Two features cut the section creating possible gaps and/or overlaps < 1 m. Feature B is a high-angle fault. Feature A is man-made: a ramp in the quarry face.

Zone at the base to the *Rotalipora ticinensis* Zone at the top. Preservation of the planktic foraminifers in the micritic pelagic limestone fluctuates throughout the sequence and improves in the younger zones as expected [*Sliter*, 1989]. Likewise, the size, abundance, and diversity of planktic foraminifers also increases upsection. Planktic foraminifers were examined in thin section based on taxonomic criteria derived from the internal and external morphologies of individual specimens. Other organisms present include rare benthic foraminifers, represented mostly by *Gavelinella* and *Gyroidinoides*, nodosariids, agglutinated species, poorly preserved radiolaria, rare *Inoceramus* prisms and fish debris. The lower half of the sequence (20.14–35.86 m) falls within the upper portion of the *Ticinella primula* Zone. The possible occurrence of *Hedbergella simplex* at 20.14 m and the first appearance of *Globigerinelloides* at 25.43 m indicates that these samples are already in the upper half of the zone. These are followed by the first appearance of *Ticinella praeticinensis* and *T. roberti* between 32.40 m and 35.86 m that occur at the top of the *T. primula* Zone. Also in this latter interval is the transition from *Biticinella subbreggiensis* to *B. breggiensis* sensu stricto that takes place at 36.48 m. The *Biticinella breggiensis* Zone thus extends from 36.48 m to 48.65 m and is subdivided into the lower *Ticinella praeticinensis* Subzone and the upper *Rotalipora subticinensis* Subzone. The boundary between the subzones is recognized by the first appearance of the genus *Rotalipora* that occurs between 40.57 m and 41.40 m. The upper boundary of the *Rotalipora subticinensis* Subzone is placed at 48.65 m on the basis of the first appearance of *Praeglobotruncana delrioensis* and the lack of *Rotalipora ticinensis*. The first occurrence of *R. ticinensis* at 50.05 m identifies the base of the *R. ticinensis* Zone.

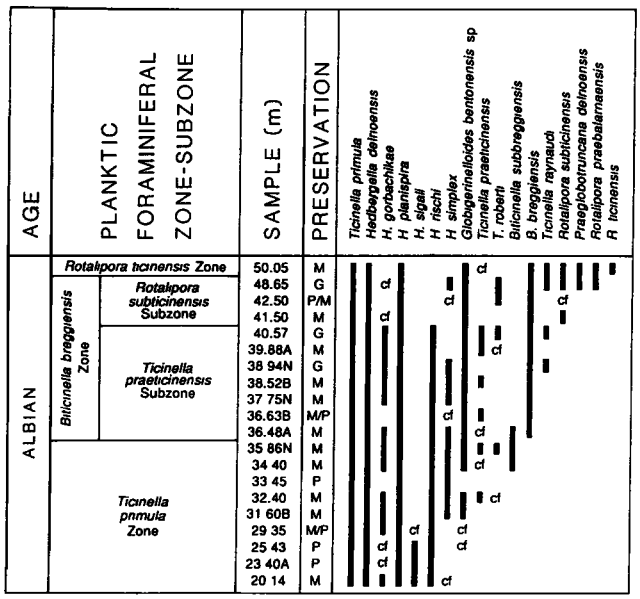


Fig. 4. Foraminiferal biostratigraphy with range chart for the Albian Contessa section. Foraminiferal zones are after *Sliter* [1989]. Preservation: G, good; M, moderate; P, poor; cf. probable identification.

Calcareous Nannofossils

Calcareous nannofossils were analyzed in 45 subsamples of unheated paleomagnetic specimens (22.77–50.05 m). Preservation is poor throughout the sampled section. Most specimens are highly overgrown and etched, and diversities are low. The poor nannofossil preservation may partially reflect a sampling bias as the paleomagnetic specimens preferentially sample the most indurated beds. To overcome the problems with preservation, smear slides were greatly concentrated in 13 select samples to detect the rarest of

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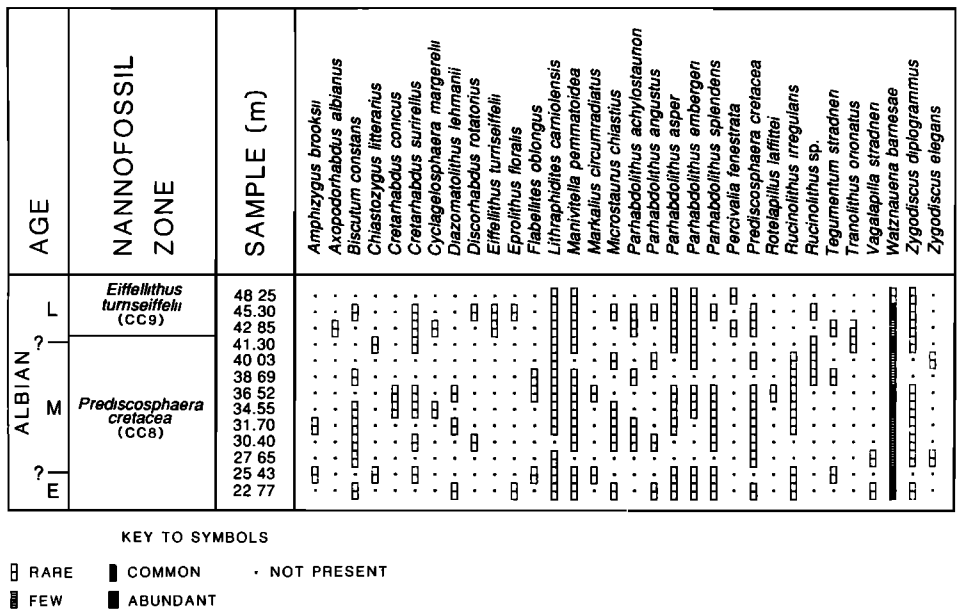


Fig. 5. Nannofossil biostratigraphy with range chart for the Albian Contessa section. Nannofossil zones after *Sissingh* [1977]. Quantification of species abundances is as follows: rare, less than five specimens observed; few, 5-10 specimens; common, 10-50 specimens; abundant, >50 specimens observed.

occurrences. These provide fairly precise correlation of the Contessa section to stratotype, parastratotype, DSDP and Ocean Drilling Program (ODP) sections and allow delineation of standard zones such as those of *Sissingh* [1977] (Figure 5).

The lower part of the section (22.77–42.85 m) lies within the *Prediscosphaera cretacea* nannofossil Zone (CC8) based on the occurrence of the nominate taxon without *Eiffellithus turriseiffelii*. Unusually common in this interval is the species *Rucinolithus irregularis*, which goes extinct towards the top of this zone (T.J. Bralower, unpublished data, 1990, DSDP site 545, ODP site 763). This taxon is constructed of seven imbricated plates but appears to evolve into a form with six imbricated plates towards the top of the section; this form is designated as *Rucinolithus sp.* The first occurrence of *Tranolithus orionatus* lies at 41.30 m, but this solution-resistant species is unusually rare. The upper part of the section (42.85–48.25 m) lies within the *Eiffellithus turriseiffelii* Zone (CC9) as indicated by the occurrence of the nominate taxon.

Nannofossil biostratigraphy of stratotype and parastratotype sections [e.g., *Thierstein*, 1973] suggests that the age of this interval ranges from the latest part of the early Albian or earliest part of the middle Albian to the earliest late Albian. Unusually close spacing of events at the top of the section (40.03–42.85 m), including the first occurrences of *T. orionatus, Axopodorhabdus albianus*, and *E. turriseiffelii* and the last occurrence of *R. irregularis*, suggests that this interval is condensed relative to the lower part of the section. These events are spread out over tens of meters at other sites including DSDP site 547 [*Wiegand*, 1984; *Leckie*, 1984] and ODP site 763 [*Bralower and Siesser*, 1991]. The order of these events is, however, comparable to other sections.

Paleontological Correlation

Planktic foraminiferal and nannofossil occurrences yield consistent ages throughout the sequence at Contessa Quarry. Both biostratigraphies indicate an age range of middle Albian for the lower sequence to late Albian for the upper half. Of special interest is the occurrence of the nannofossil *Eiffellithus turriseiffelii* in the foraminiferal *Rotalipora subticinensis* Subzone in samples 42.85 and 45.30. This co-occurrence prior to the first appearance of *R. ticinensis* represents an earlier record of *E. turriseiffelii* than that previously documented [e.g., *van Hinte*, 1976; *Tornaghi et al.*, 1989]. The presence of abundant, well-preserved foraminifers in the limestone samples (Figure 4) eliminates selective preservation as the primary cause of this unusual occurrence. The record is more likely due to the close sample spacing and continuous depositional sequence at the Contessa Quarry section. For example, one of the best dated coeval sequences was recovered from the North Atlantic at DSDP sites 545 and 547, but an unconformity extends across the critical interval within the *Ticinella praeticinensis* Subzone of the *Biticinella breggiensis* Zone to the overlying *Rotalipora ticinensis* Zone [*Leckie*, 1984].

PALEOMAGNETIC AND ROCK MAGNETIC ANALYSES

Natural remanent magnetizations (NRMs) from the Contessa section are distinct from the axial geocentric dipole field direction at the site (Figure 6). Detailed thermal demagnetization of the NRM shows that the unblocking spectrum of individual samples varies systematically through the section. In general, NRM intensity decreases much more rapidly with increasing demagnetization temperature at the base and top of the section than the middle portion (Figure 7). One way to quantify this variability is to plot the demagnetization temperature ($T_{0.2NRM}$) at which the magnetic intensity falls to 20% of the initial NRM intensity (Figure 8). Using this temperature as an attribute for division, the following three intervals can be defined: (1) low-unblocking temperature zone, 20.14-30.97 m, $T_{0.2NRM}$ = 330°C, σ = 10°C, (2) high-unblocking temperature zone, 31.23-38.14 m, $T_{0.2NRM}$ = 520°C, σ = 130°C, and (3) intermediate-unblocking temperature zone, 38.20-50-.05 m, $T_{0.2NRM}$ = 390°C, σ = 50°C. These intervals range from

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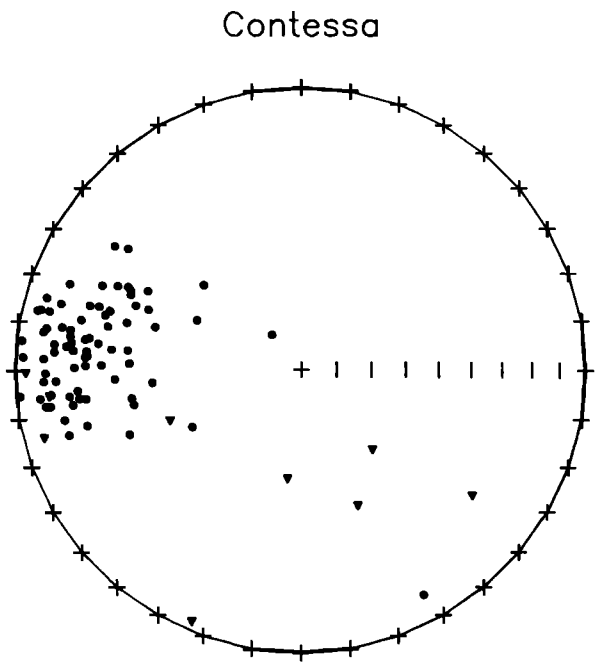


Fig. 6. Stereographic projection of natural remanent magnetizations from the Albian Contessa section. Solid circles, positive inclinations; solid triangles, negative inclinations.

relatively uniform low-unblocking temperatures to generally high-unblocking temperatures with great variability. Even this threefold division, however, simplifies the variability. For example, the middle zone (interval 2) consists of alternating intervals with $T_{0.2NRM}$ of approximately 350°C and intervals with $T_{0.2NRM}$ >500°C. Such changes over relatively short stratigraphic distances are unlike the optimal magnetic behavior in other Umbrian sections [*Lowrie and Heller*, 1982] which has been central to the success of magnetostratigraphic studies. A more detailed rock magnetic analysis of the samples, therefore, was conducted.

The acquisition of isothermal remanent magnetization (IRM) is a common technique aiding the identification of ferromagnetic mineral content. Interpretation of IRM acquisition curves, however, can be ambiguous since some ferromagnetic minerals, such as goethite and hematite, can have similar coercivities [*Heller*, 1978; *Lowrie and Heller*, 1982]. Stepwise thermal demagnetization of an IRM can resolve such ambiguities since many minerals which have similar maximum coercivities have different unblocking temperatures [*Heller*, 1978]. Recently, a modification of the thermal demagnetization of IRM has been presented providing even greater resolution of ferromagnetic mineral content, especially in relatively weakly magnetized pelagic limestones [*Lowrie*, 1990]. The different coercivity fractions of a rock are remagnetized by an IRM acquired in successively smaller fields along orthogonal directions. The stepwise thermal demagnetization of each othogonal component is plotted separately enabling both the recognition of ferromagnetic minerals having overlapping maximum coercivities and some idea of the relative contributions of the various coercivity fractions. We have applied this method to selected subsamples of the paleomagnetic cores and hand samples collected at the Contessa Quarry. Samples were prepared as 2-3 cm^3 cubes. Each of the unblocking temperature zones is discussed below with regard to their paleomagnetic components and likely magnetic mineralogy.

Interval 1, Low NRM Unblocking Temperatures

Initial IRM acquisition experiments on samples from 20.14 m and 21.00 m to 1 T suggested a relatively restricted coercivity distribution (Figure 9*a*). Therefore, an IRM was imparted along only two axes. Subsequent thermal demagnetization of the coercivity fraction magnetized at 0.15 T shows a maximum unblocking temperature near 580°C, indicating the presence of magnetite (Figure 9*b*). Unblocking temperatures extending above 600°C in the

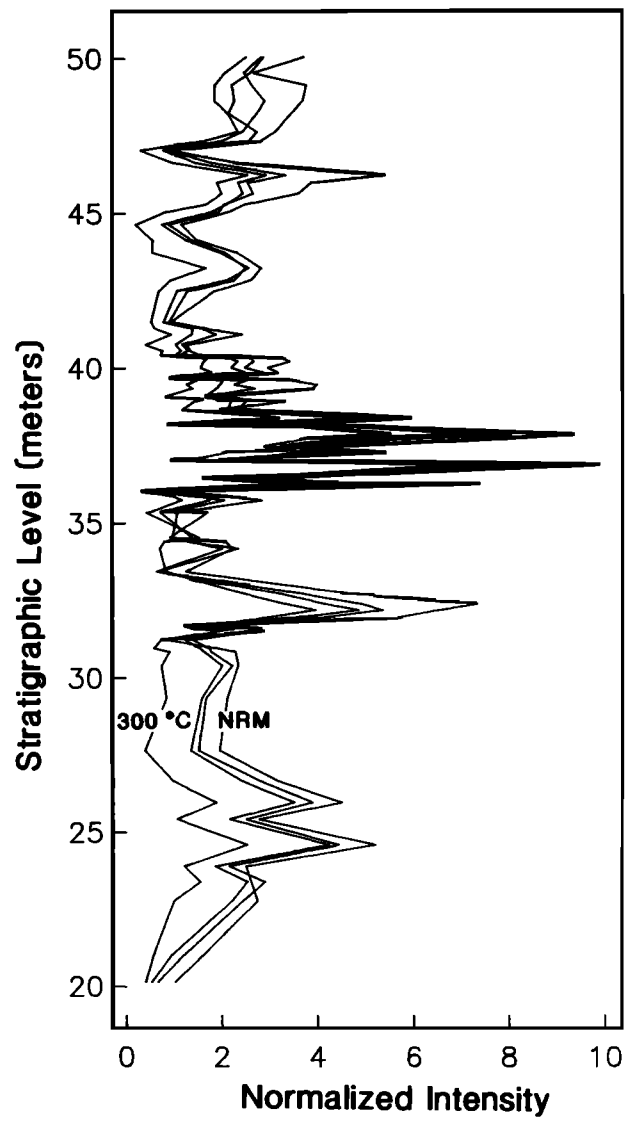


Fig. 7. Differences in the decay of NRM intensity along the Albian Contessa section with progressive thermal demagnetization. Intensity of initial natural remanent magnetization (NRM) and NRM intensity remaining after thermal demagnetization at 100°C, 200°C, and 300°C shown normalized as lines of constant demagnetization temperature versus stratigraphic meter level. Spacing between adjacent lines represents intensity removed during the appropriate temperature increment; differences in this spacing occur with stratigraphic meter level reflecting differences in rock magnetic content through the measured section.

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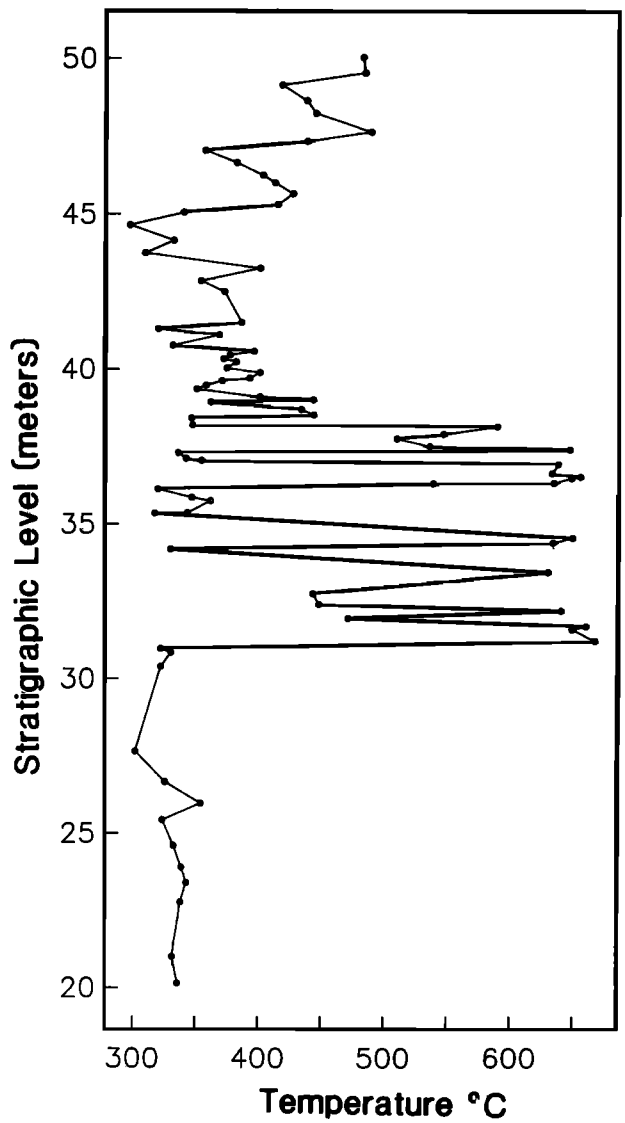


Fig. 8. Thermal demagnetization temperature at which the remaining NRM intensity equals 0.2 of the initial natural remanent magnetization intensity versus stratigraphic level (m) for the Albian Contessa section.

thermal demagnetization of the coercivity fraction magnetized at 1 T indicates that some hematite is also present in these rocks. The relative quantities of these minerals appear to change stratigraphically up the section toward interval II. At 23.40 m, unblocking temperatures above 600°C indicate the presence of hematite in the coercivity fractions magnetized in fields of 0.15, 0.6, and 5 T (Figure 9). Thermal demagnetization of the coercivity fraction magnetized at 0.15 T may indicate the presence of a small amount of magnetite with a distributed unblocking temperature range mirroring a distributed grain size. The dominant magnetic mineral, however, appears to be hematite.

Thermal demagnetization of the NRM of rocks from interval I first removes a distinct component of magnetization between 50°C and 150°C (Figure 9*d*). Although both polarities are displayed by this component, the predominant polarity is reversed. Detailed demagnetization of this component shows an in situ direction close to a Matuyama chron (2.48-.73 Ma) field direction. After removal of this low-unblocking temperature component, most samples displayed single-component behavior of normal polarity. After tilt correction, this component is similar in direction to that derived from other paleomagnetic studies of Umbrian Cretaceous rocks. We believe this component is carried by magnetite in most of interval I and by magnetite and hematite as one approaches interval II. Directions became erratic in many samples at temperatures ranging from 350°C to 475°C (Figure 9*d*). In these samples magnetic susceptibility values begin to increase at 475°C and thereafter rise sharply with increasing temperature (Figure 9*c*). The increase in magnetic susceptibility signals the growth of a new mineral, which is most likely magnetite. The observation that directional instability occurs at slightly lower temperatures than the increase in magnetic susceptibility in some samples suggests that NRM stability may be a more sensitive gauge of initial alteration during thermal demagnetization in these rocks.

An exception to the general demagnetization behavior outlined above for interval I was noted at 29.35 m. At demagnetization temperatures to 575°C, the resultant vector displayed a trajectory away from the origin suggesting the presence of a reversed polarity magnetization with higher unblocking temperatures. Although the directions became unstable at 600°C and this component could not be fully resolved, the similarity of this demagnetization behavior to samples from interval II, discussed below, leads us to suggest the presence of a reversed polarity component of magnetization at 29.35 m carried by hematite.

Interval II, High NRM Unblocking Temperatures

Initial IRM acquisition experiments on specimens from interval II showed evidence for a high coercivity mineral which required fields greater than 1 T for saturation (Figure 10*a*). An IRM was imparted along three orthogonal axes in fields of 0.15, 0.6, and 5 T. Thermal demagnetization of these components all showed evidence for hematite as evidenced by maximum unblocking temperatures above 600°C (Figure 10*b*). The intensity of the IRM acquired at 5 T varied by more than an order of magnitude within interval II. As in the samples from interval I, small amounts of magnetite may also be present in the coercivity fraction magnetized at 0.15 T of certain specimens (e.g., Figure 10, sample 38.94) as indicated by distributed unblocking temperature spectra. Since magnetite has a maximum coercivity of 0.3 T, the contribution of magnetite grains having coercivities between 0.15 T and 0.3 T would also be present in the coercivity fraction magnetized at 0.6 T. The coercivity fraction magnetized at 0.6 T also has a distributed unblocking temperature range for certain specimens (e.g., Figure 10, sample 38.94). The presence of small amounts of magnetite in the coercivity fractions magnetized at 0.15 and 0.6 T is difficult to distinguish from hematite with a range of grain sizes and hence unblocking temperatures. If present, however, the amount of magnetite must be small relative to hematite because the rapid saturation expected for magnetite is not obvious in the initial IRM acquisition curves.

Thermal demagnetization of the NRM from interval II again removed low-unblocking temperature components with directions close to the present-day field (but too steep) and the Matuyama chron field (Figure 10*d*). The "Matuyama" direction occasionally persisted to 300°C or higher. (In some samples a hint of the "Matuyama" component is seen

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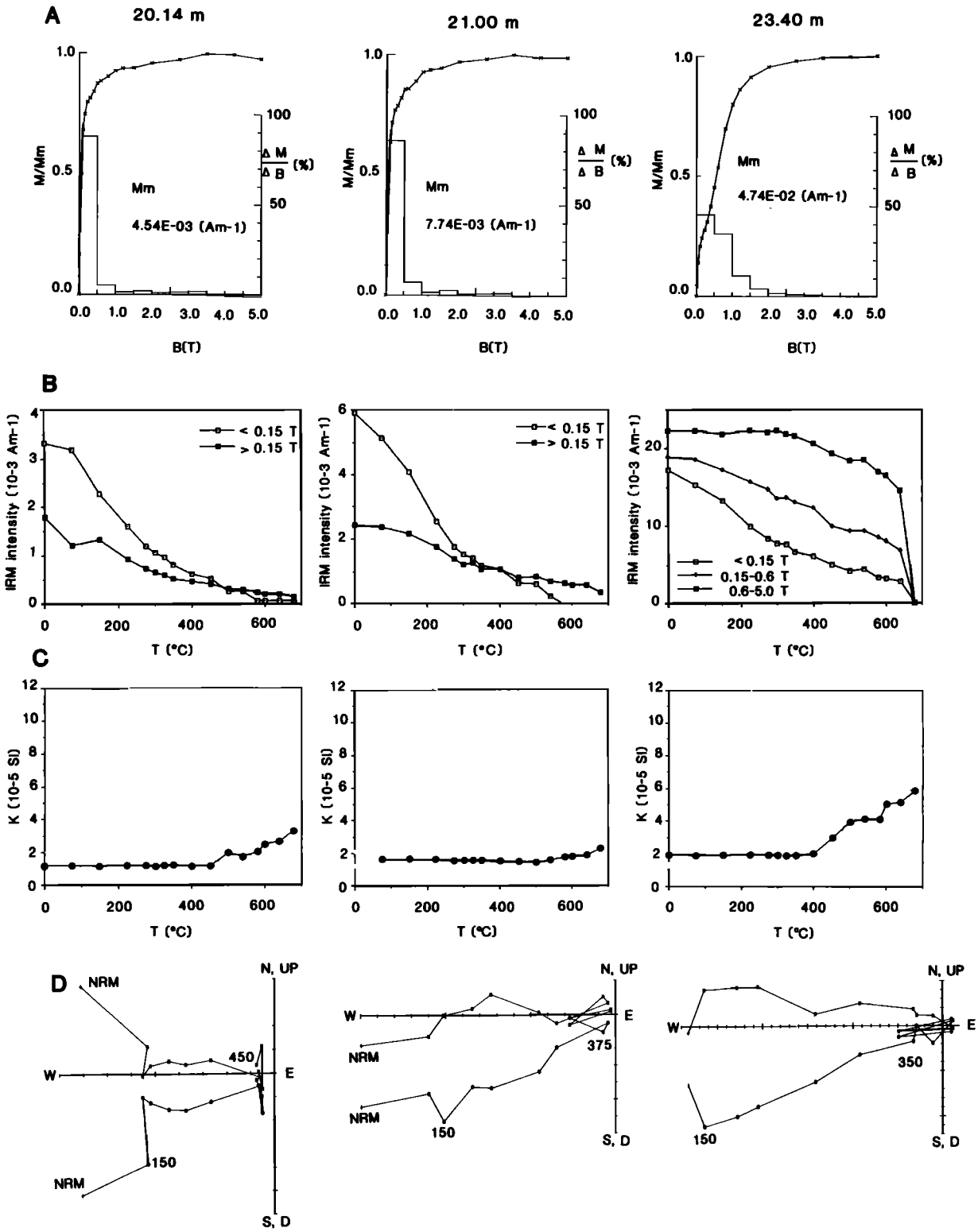


Fig. 9. Rock magnetic data from characteristic samples of rock magnetic interval I (20.14-30.97 m) of the Albian Contessa section. (*a*) Progressive acquisition of an isothermal remanent magnetization in fields to 5 T. (*b*) Thermal demagnetization of a composite IRM acquired along two orthogonal axes in fields of 0.15 T and 5 T (samples 20.14 and 21.0) and along three orthogonal axes in fields of 0.15 T, 0.6 T, and 5 T (23.40 m). (*c*) Volume magnetic susceptibility versus thermal demagnetization temperature for specimens used in IRM acquisition experiments (Figure 9*b*). (*d*) Orthogonal vector plots of progressive thermal demagnetization of natural remanent magnetization, shown in geographic (in situ) coordinates. Circles, inclination; crosses, declination. Demagnetization steps (in degrees Celsius) shown: 20.14 m, 0, 100, 150, 200, 250, 300, 325, 350, 375, 400, 425, 450; 21.00 m, 0, 100, 150, 200, 250, 300, 325, 350, 375, 400, 425; 23.40 m, 100, 150, 200, 250, 300, 325, 350, 375, 400, 425, 450, 475.

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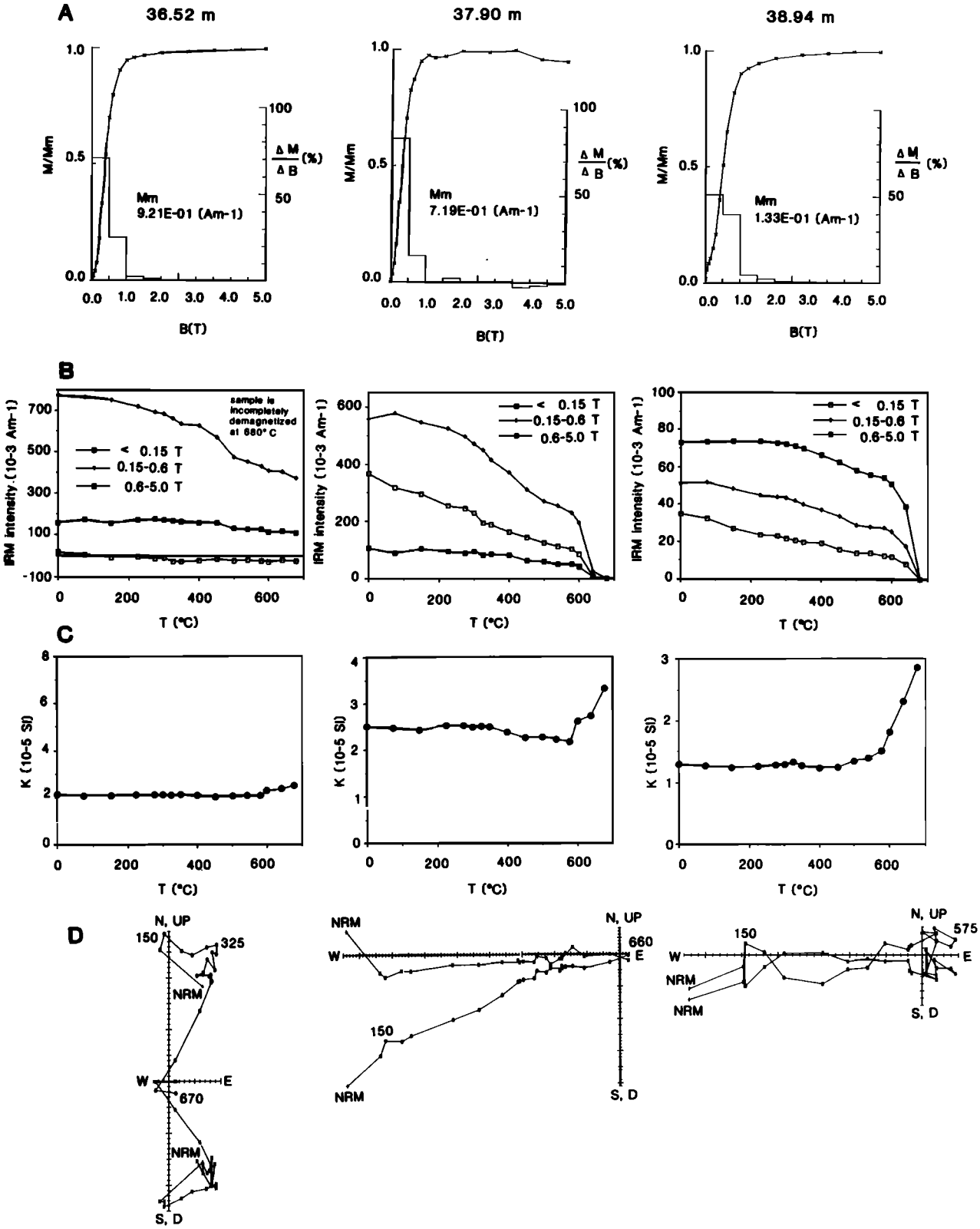


Fig. 10. Rock magnetic data from characteristic samples of rock magnetic interval II (31.23-38.14 m) of the Albian Contessa section. (*a*) Progressive acquisition of an isothermal remanent magnetization in fields to 5 T. (*b*) Thermal demagnetization of a composite IRM acquired along three orthogonal axes in fields of 0.15 T, 0.6 T, and 5 T. (*c*) Volume magnetic susceptibility versus thermal demagnetization temperature for specimens used in IRM acquisition experiments (Figure 10*b*). (*d*) Orthogonal vector plots of progressive thermal demagnetization of natural remanent magnetization, shown in geographic (in situ) coordinates. Circles: inclination; crosses: declination. Demagnetization steps (in degrees Celsius) shown: 36.52 m, 0, 100, 150, 200, 250, 300, 325, 350, 375, 400, 425, 450, 475, 500, 525, 550, 575, 600, 625, 640, 650, 660, 670; 37.90 m, 0, 100, 150, 200, 250, 300, 325, 350, 375, 400, 425, 450, 475, 500, 525, 550, 565, 600, 615, 625, 640, 650, 660; 38.94 m, 0, 100, 150, 200, 250, 300, 325, 350, 375, 400, 425, 450, 475, 500, 525, 550, 575, 600, 625, 640, 650. A reversed polarity characteristic magnetization is isolated above 600°C in samples 36.52 and 38.94 but with different intensities relative to other components in the samples. Sample 37.90 carries a normal polarity ChRM at unblocking temperatures above 600°C.

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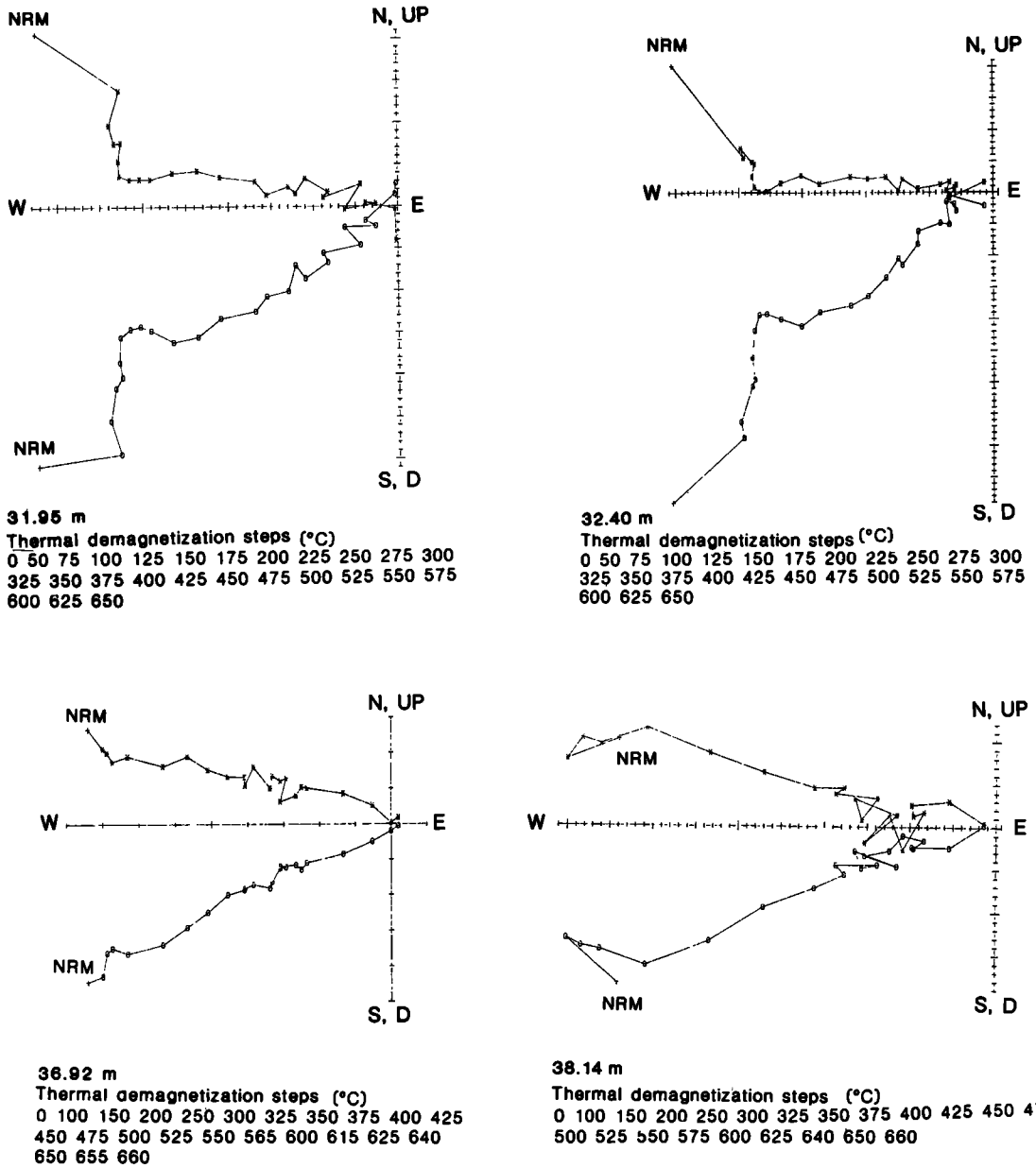


Fig. 11. Normal polarity characteristic directions isolated at high unblocking temperatures. Circles, inclination; crosses, declination.

at substantially higher unblocking temperatures; e.g., Figure 10*d*, 36.52 m, 325-400°C). Since the unblocking temperature range of these directions extends above the Néel point of goethite, a fine-grained hematite carrier is inferred.

After removal of the low unblocking temperature magnetization, a normal polarity direction similar to that observed in other studies of Cretaceous Umbrian rocks was isolated in all samples. Substantial differences were observed, however, in the maximum unblocking temperature at which this component was defined. In some samples, this component extended to temperatures as high as 650°C, while in other samples directions became erratic at temperatures above 475°C, similar in behavior to some samples from interval I. In one zone (32.75-35.36 m) the normal polarity component did not trend to the origin suggesting the presence of an additional unresolved component. Based on our IRM experiments, we conclude that the normal polarity component is carried primarily in hematite with a possible minor contribution due to magnetite.

Some samples from interval II also contain a reversed polarity characteristic magnetization which was isolated (i.e., demagnetized) above 600°C, after first removing the normal polarity magnetization (Figure 10*d*). This reversed polarity component did not demagnetize completely until temperatures as high as 680-690°C. Although this reversed polarity characteristic magnetization is only isolated at very high unblocking temperatures, this component was not observed at each horizon. Instead, the normal polarity component is well-defined at very high unblocking temperatures in certain beds (Figures 10 and 11).

Lateral Persistence of Reversed Characteristic Directions

To check whether the high unblocking temperature reversed polarity magnetizations are persistent laterally, two beds from

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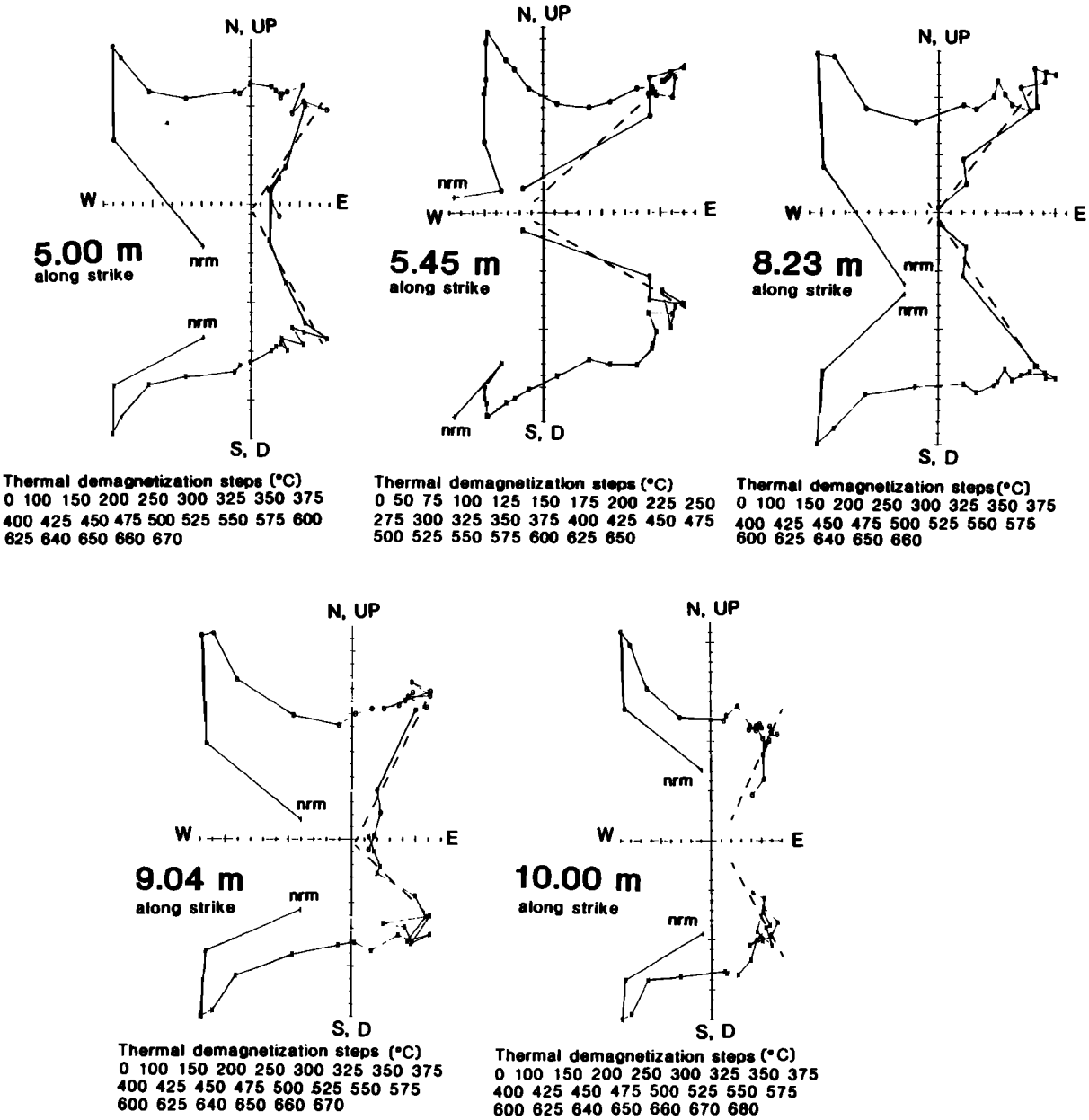


Fig. 12. Orthogonal vector plots of thermal demagnetization data for samples from bed A (36.48 m). Data are shown in geographic (in situ) coordinates. Circles, inclination; crosses, declination. Meter levels (bold) represent horizontal distance along the bed. The dashed lines represent least squares line fits to the demagnetization data.

interval II which displayed these magnetizations were sampled for distances up to ~7 m along strike. Demagnetization of samples from both bed A (36.48 m) and bed B (35.86 m) show that the reversed polarity magnetizations are persistent. (Figures 12-13) (Tables 4-5 and 10). As in specimens from the other horizons displaying the reversed magnetizations, thermal demagnetization (above 250°C) first removed a normal polarity direction prior to removing the high-temperature reversed polarity direction. Samples from bed A displayed consistency in directions, unblocking temperature ranges, and relative intensities of the normal and reversed polarity high-temperature components along strike. Demagnetization of these samples also demonstrate that the reversed polarity component of magnetization is never clearly defined until temperatures above 575–600°C, close to the Curie point of magnetite. Samples from bed B (Figure 13) showed consistency in directions and unblocking temperature ranges but showed more variability in the relative intensities of the normal and reversed polarity components than bed B.

Alternating Field Demagnetization

To examine further the magnetic carriers in beds A and B, several samples were selected for alternating field demagnetization (Figure 14). These experiments reveal that the Brunhes field component and the normal polarity component are carried at least in part by a mineral that can be demagnetized in peak alternating fields less than 80 mT. The obvious candidate for such a mineral is a titanomagnetite, possibly end-member magnetite. The presence of small amounts of magnetite also accounts for the removal of the normal polarity component at peak thermal demagnetization

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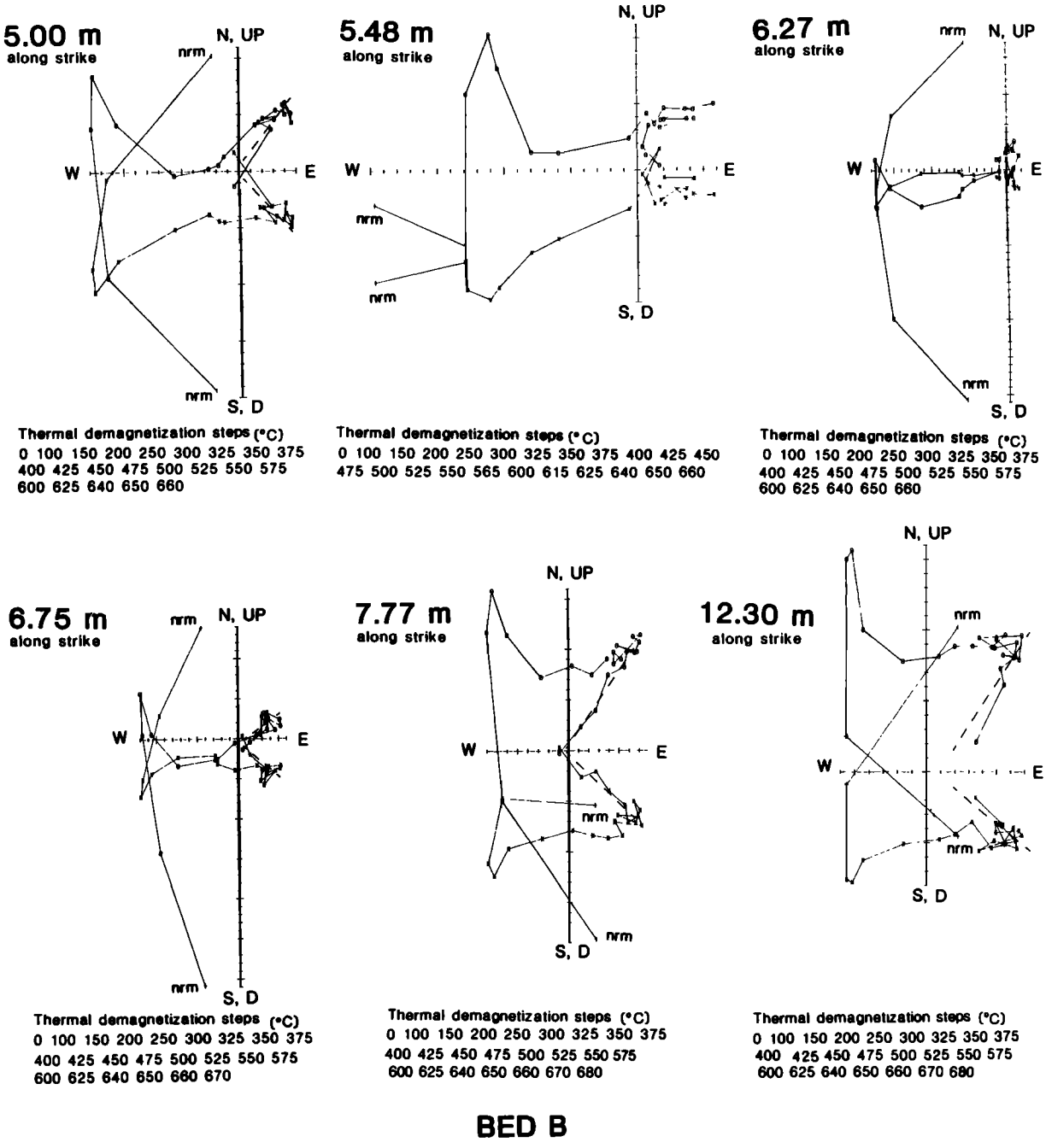


Fig. 13. Orthogonal vector plots of thermal demagnetization data for samples from bed B (35.86 m). Data are shown in geographic (in situ) coordinates. Circles, inclination; crosses, declination. Meter levels (bold) represent horizontal distance along the bed. The dashed lines represent least squares line fits to the demagnetization data. Characteristic remanent magnetizations for samples MF35.86 and B6.27 estimated from Fisherian averages of high-temperature data (see Table 5).

temperatures close to the Curie point of magnetite in samples from beds A and B.

Interval III, Intermediate NRM Unblocking Temperatures

As in some samples from interval I, initial IRM acquisition experiments on samples from interval III (Figure 15*a*) showed a much more discrete range of coercivities as compared to those of interval II. Thus, an IRM was imparted along only two directions and subsequently demagnetized. Maximum unblocking temperatures near 580°C in both coercivity fractions in a sample from 54.40 m indicate the presence of magnetite (Figures 15*b*). A small amount of hematite may be present in a sample from 53.00 m, but again the dominant magnetic mineral appears to be magnetite on the basis of the unblocking temperature spectrum.

Both polarities of the low-unblocking temperature magnetizations are often clearly defined in a single sample (Figure 15*d*). In such samples the normal polarity magnetization close to the present-day field is always carried in the lower unblocking temperature range, between 100° and 150°C. The reversed polarity direction close to the Matuyama chron field

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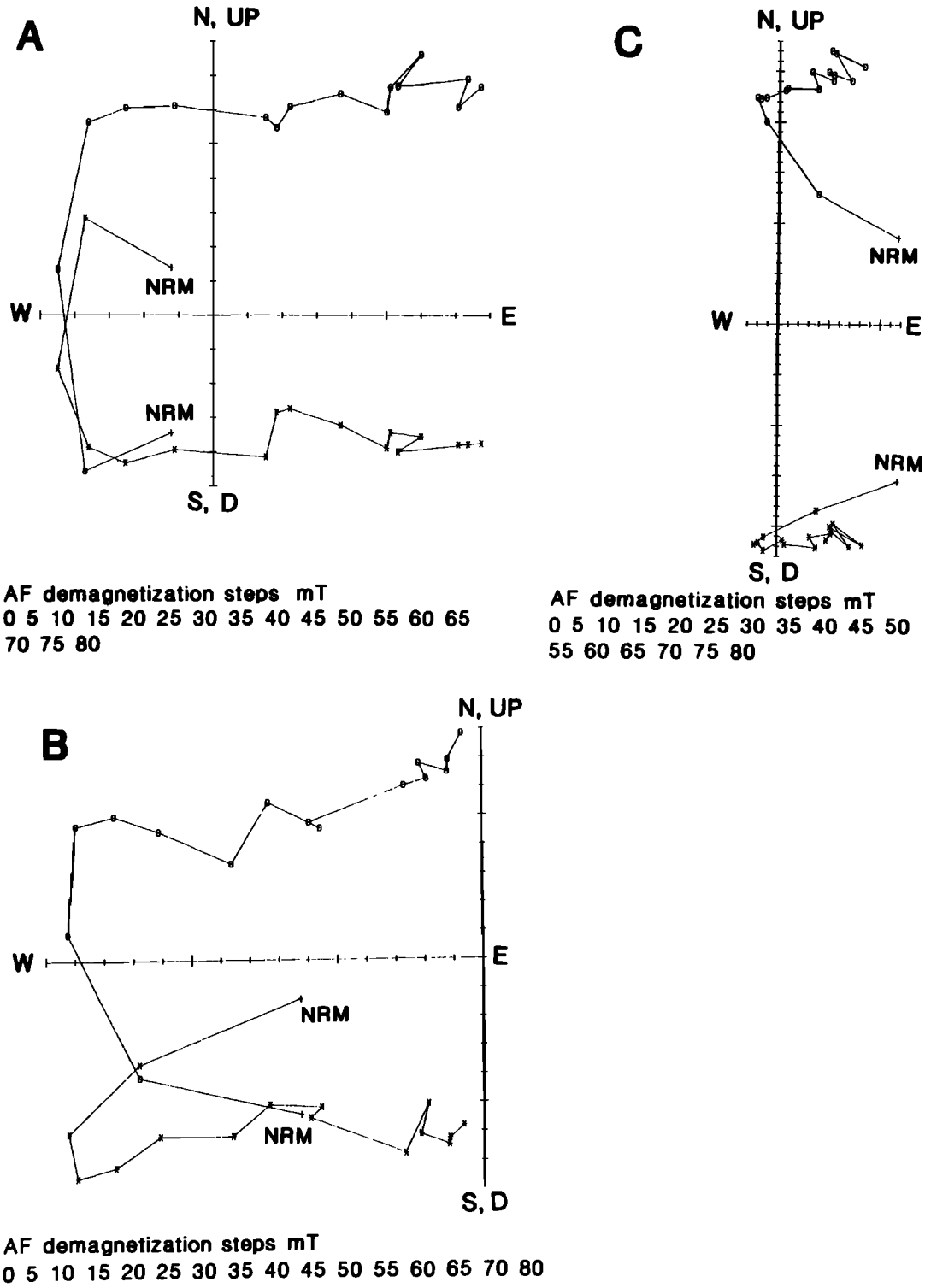
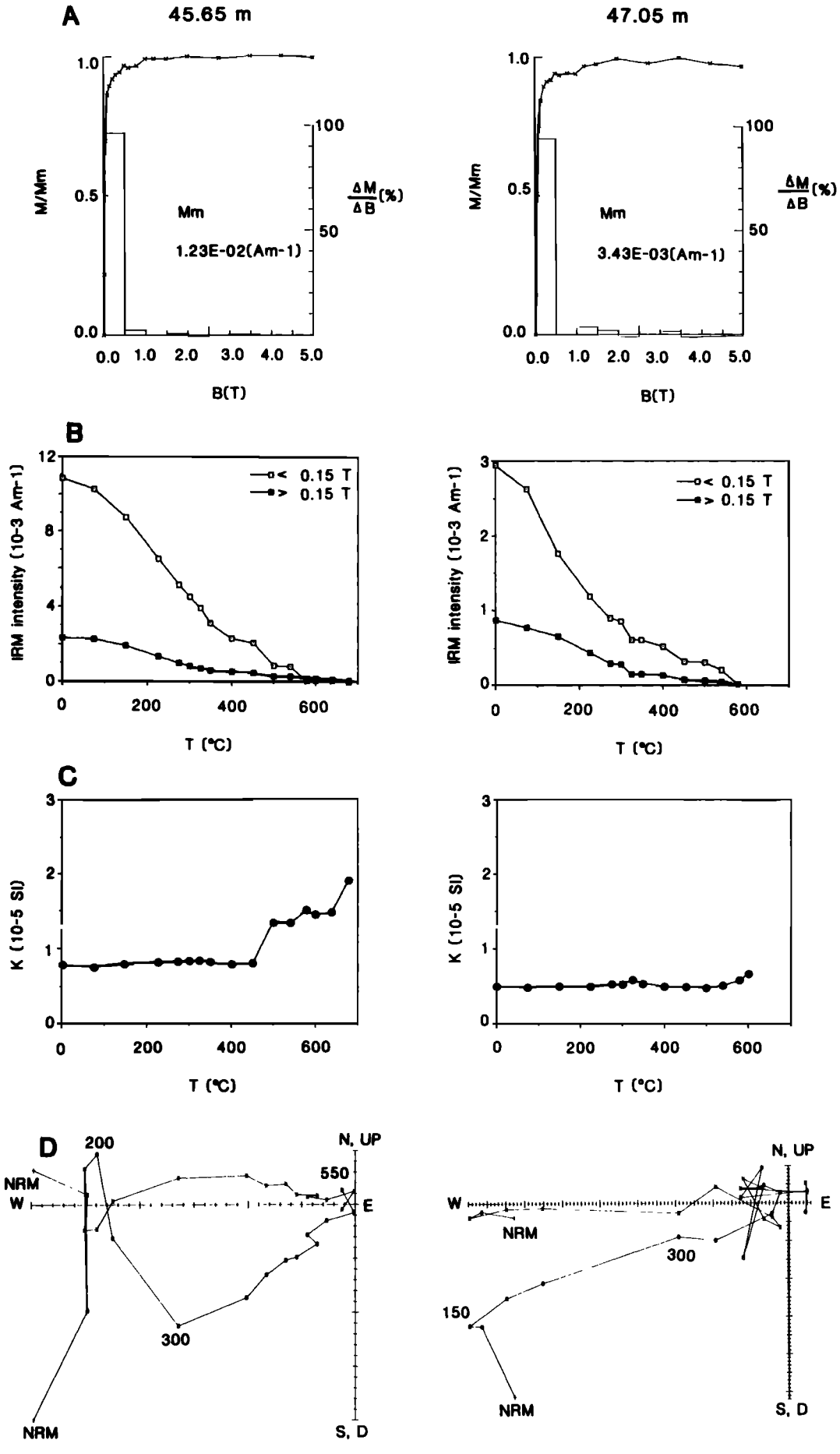


Fig. 14. Orthogonal vector plots of progressive alternating field demagnetization. All data shown in geographic (in situ) coordinates. Circles, inclination; crosses, declination. (*a*) Sample from bed B (35.86 m) at a horizontal distance of 7.77 m along the bed. (*b*) Sample from bed B (35.86 m) at a horizontal distance of 12.05 m along the bed. (*c*) Samples from bed A (36.48 m) at a horizontal distance of 6.55 m along the bed.

direction is isolated between 150° and 300°C. The component of magnetization close to the present-day field is most likely carried mainly in a combination of coarse magnetite and fine-grained hematite, while the Matuyama component may be carried mainly by fine-grained hematite.

Above 300°C, single-component behavior of normal polarity was dominant, decaying to near zero intensity at 575°C. Two exceptions to this behavior were noted. First, a zone (39.10-39.88 m) was observed where the normal component did not trend to the origin. Second, high unblocking

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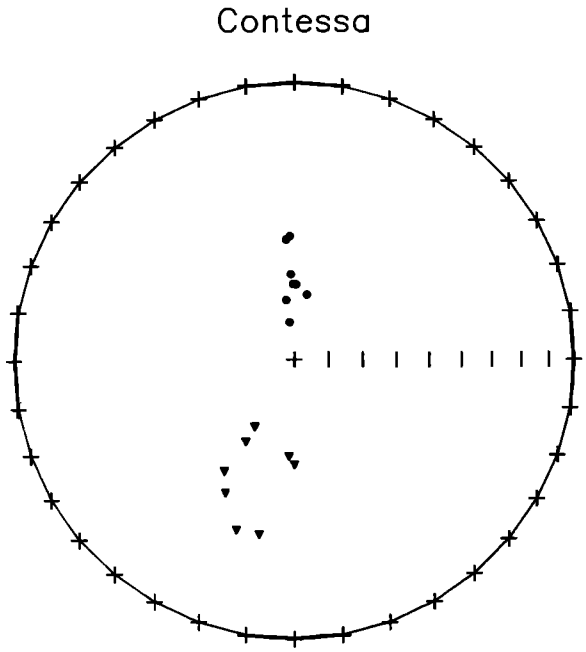


Fig. 16. Stereographic projection of components isolated at low unblocking temperatures from selected samples of the Albian Contessa section shown in geographic (in situ) coordinates. Positive inclinations (solid circles) are interpreted as a Brunhes field overprint (Table 1); negative inclinations (filled triangles) are interpreted as a "Matuyama" (or late Tertiary) field overprint (Table 2).

temperature components (>575°C) of reversed polarity were observed from some horizons near interval II. As opposed to interval II, this reversed component appears as a much less intense component compared to the normal polarity component. As in interval II, this reversed polarity component was clearly defined (i.e., demagnetized) only after the removal of a normal polarity component which extended to unblocking temperatures as high as 575°C. In contrast to interval II, we believe this high unblocking temperature reversed polarity component is carried by hematite.

SUMMARY OF PALEOMAGNETIC AND ROCK MAGNETIC DATA

Brunhes Field Direction

All three stratigraphic intervals within the section contain a low unblocking temperature component of NRM with in situ direction close to the Brunhes chron (0.73–0 Ma) field (Figure 16). Detailed thermal demagnetizations (25°C steps) were applied to only select samples in the lowest unblocking temperature range (0–200°C). The inclination of the component removed during this detailed demagnetization was fit by least squares line fits. The fact that this component is often steeper than the Brunhes field perhaps indicates its incomplete separation from higher stability components (Table 1). The likely mineral carriers are fine-grained hematite and magnetite. The magnetization mechanism may be a viscous remanent magnetization (VRM).

TABLE 1. Brunhes Overprint Directions in Geographic Coordinates

Sample (m)	Temperature Range, °C	n	D, deg	I, deg
MF(31.60)	100.0-175.0	4	357.9	53.4
MF(31.95)	100.0-175.0	4	356.0	54.4
MF(32.40)	100.0-175.0	4	357.6	64.9
MF(36.48)	100.0-175.0	4	352.2	72.4
MF(37.05)	125.0-175.0	3	352.9	79.0
MF(40.33)	75.0-175.0	5	1.5	67.9
MF(39.10)	75.0-175.0	5	11.0	70.6
MF(39.48)	75.0-175.0	5	358.9	67.8

In sample (m) the (m) indicates meter level in the section; n, number of points used in fit; D, declination; I, inclination.

Matuyama (or Late Tertiary) Field Direction

All three stratigraphic intervals also contain evidence of a component of NRM with in situ direction parallel to the Matuyama field direction (Figure 16) (Table 2). It is also possible, however, that this component represents a magnetization acquired in an older reversed polarity chron of the late Tertiary. The unblocking temperature range over which this component is isolated varies slightly throughout the section. At the base (intervals I) the component is sometimes carried in the same unblocking temperature range as that carrying the Brunhes field direction. In the rest of the section, the "Matuyama" component is isolated at higher unblocking temperatures, generally between 150° and 300°C. Fine-grained hematite is postulated as the mineral carrier and the origin of magnetization is tentatively identified as VRM. If the "Matuyama" component is also carried by higher unblocking temperature hematite, as suggested by a few samples (e.g., Figure 10*d*, 36.52 m, 325-400°C), an origin by chemical remanent magnetization is more likely.

TABLE 2. Matuyama Overprint Directions in Geographic Coordinates

Sample (m)	Temperature Range, °C	n	D, deg	I, deg
MF(27.65)	75.0-125.0	3	207.5	-45.7
MF(30.40)	75.0-125.0	3	191.4	-36.7
MF(33.45)	75.0-125.0	3	198.9	-36.2
MF(49.55)	150.0-250.0	3	183.1	-62.0
MF(50.05)	150.0-300.0	4	179.9	-59.5
MF(48.25)	200.0-300.0	3	210.9	-62.5
MF(47.35)	200.0-300.0	3	210.9	-67.7
MF(39.10)	175.0-250.0	4	212.3	-51.4

See Table 1 footnotes.

Fig. 15. Rock magnetic data from characteristic samples of rock magnetic interval III (38.20-50.05 m) of the Albian Contessa section. (*a*) Progressive acquisition of an isothermal remanent magnetization in fields to 5 T. (*b*) Thermal demagnetization of a composite IRM acquired along two orthogonal directions in fields of 0.15 T and 5 T. (*c*) Volume magnetic susceptibility versus thermal demagnetization temperature for specimens used in IRM acquisition experiments (Figure 15*b*). (*d*) Orthogonal vector plots of progressive thermal demagnetization of natural remanent magnetization shown in geographic (in situ) coordinates. Circles, inclination; crosses, declination. Demagnetization steps (in degrees Celsius) shown: 45.65 m, 0, 100, 150, 200, 250, 300, 350, 375, 400, 425, 450, 475, 500, 525, 550; 47.05 m, 0, 100, 150, 200, 250, 300, 350, 375, 400, 425, 450, 475, 500, 525, 550.

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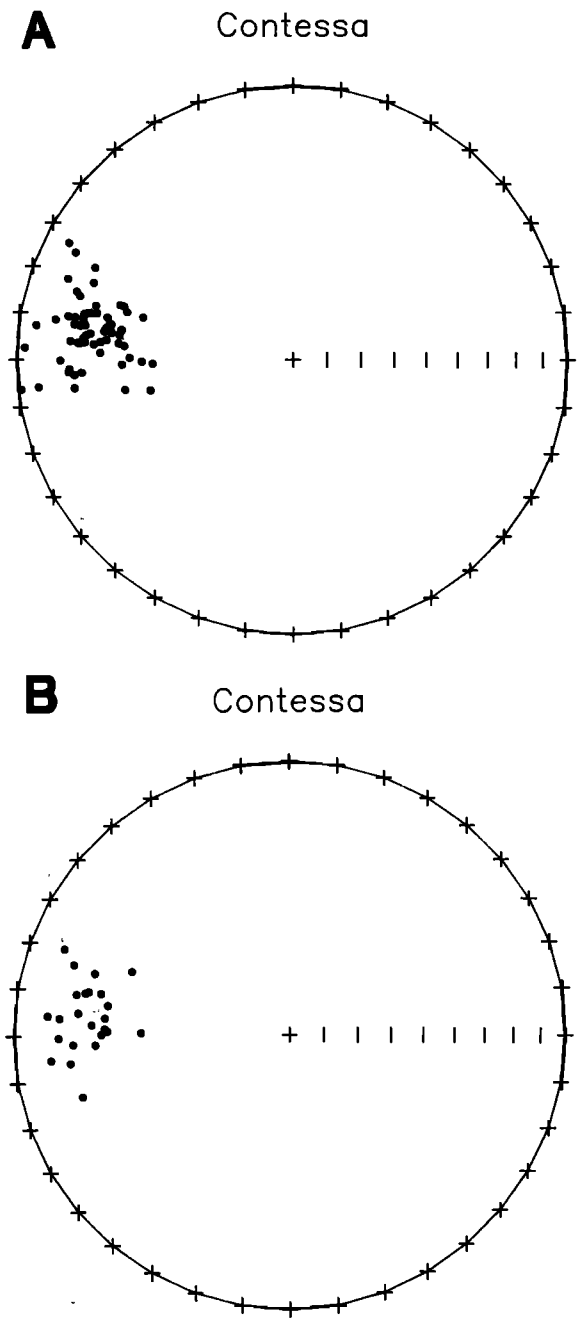


Fig. 17. Stereographic projection of normal polarity components isolated from the Albian Contessa section shown in geographic (in situ) coordinates. (*a*) Normal polarity characteristic remanent magnetizations (ChRM) from samples containing only a normal polarity ChRM (N=61, Table 3). (*b*) Normal polarity directions obtained from samples containing a reversed polarity ChRM isolated at unblocking temperatures above 600°C (Table 7).

TABLE 3. Normal Characteristic Directions in Geographic Coordinates

Sample (m)	Temperature Range, °C	n	D, deg	I, deg
MF(20.14)	300.0-350.0+O	4	279.2	21.1
MF(21.00)	250.0-350.0+O	5	275.2	30.8
MF(22.77)	200.0-400.0+O	8	279.8	32.3
MF(23.40)	250.0-350.0+O	5	279.1	23.0
MF(23.90)	300.0-400.0+O	6	279.4	24.9
MF(24.60)	300.0-350.0+O	4	278.4	34.2
MF(25.43)	200.0-300.0+O	4	274.3	28.6
MF(25.97)	250.0-550.0+O	13	285.4	27.1
MF(26.66)	200.0-400.0	7	287.6	36.2
MF(27.65)	200.0-425.0+O	10	276.0	32.7
MF(30.40)	225.0-325.0+O	6	282.3	23.9
MF(30.85)	250.0-350.0+O	5	280.6	24.0
MF(30.97)	250.0-350.0+O	6	283.5	28.0
MF(31.95)	275.0-625.0+O	16	279.4	33.2
MF(32.20)	200.0-550.0	13	263.6	1.4
MF(32.40)	275.0-575.0+O	14	274.6	38.8
MF(32.75)†‡	300.0-565.0	12	267.5	20.3
MF(35.35)†‡	250.0-350.0	4	279.3	32.0
MF(35.36)†‡	250.0-350.0	4	287.7	19.2
MF(36.30)	300.0-600.0	13	274.5	23.9
MF(36.32)	300.0-670.0	18	266.7	19.9
MF(36.63)	300.0-625.0	15	263.7	8.3
MF(36.92)	250.0-650.0+O	19	283.0	25.8
MF(37.40)	300.0-575.0	12	277.7	7.1
MF(37.50)	300.0-660.0	18	266.4	24.5
MF(37.75)	300.0-660.0	19	274.8	20.4
MF(37.90)	200.0-660.0	20	265.9	21.9
MF(38.14)	300.0-600.0	13	291.3	23.9
MF(38.52)	300.0-625.0	14	274.5	23.0
MF(39.10)‡	300.0-400.0	5	259.7	38.3
MF(39.36)‡	300.0-400.0	5	287.7	34.8
MF(39.48)‡	325.0-450.0	6	286.1	37.7
MF(39.88)‡	325.0-450.0	6	269.3	44.5
MF(40.23)	325.0-550.0+O	11	270.7	40.6
MF(40.33)	300.0-550.0+O	12	275.4	36.8
MF(40.45)	325.0-425.0+O	6	278.7	31.6
MF(40.57)	300.0-500.0+O	10	272.0	30.9
MF(40.77)	250.0-350.0+O	5	281.1	33.6
MF(41.10)	325 0-425.0	5	277.4	27.1
MF(41.30)	250.0-375.0	5	285.9	43.1
MF(41.50)	350.0-400.0+O	4	282.9	31.8
MF(42.50)	300.0-550.0	10	272.6	3.3
MF(42.85)	250.0-400.0	5	290.0	14.7
MF(43.25)	300.0-550.0+O	11	274.8	25.5
MF(43.75)	250.0-350.0	3	283.0	25.3
MF(44.15)	250.0-350.0	3	296.5	13.4
MF(44.65)	250.0-425.0	6	286.8	20.8
MF(45.05)	250.0-375.0	4	297.8	9.5
MF(45.30)	300.0-550.0+O	11	279.7	14.0
MF(45.65)	300.0-550.0+O	11	280.0	37.3
MF(46.00)	300.0-550.0+O	11	276.8	25.6
MF(46.25)	350.0-550.0	9	281.2	20.9
MF(46.65)	300.0-525.0+O	10	295.2	22.2
MF(47.05)	250.0-375.0+O	5	269.8	17.0
MF(47.35)	350.0-550.0+O	10	281.1	18.4
MF(47.65)	350.0-550.0+O	10	278.1	36.8
MF(48.25)	350.0-525.0+O	9	268.3	38.2
MF(48.65)	350.0-525.0	8	257.8	46.2
MF(49.15)	350.0-500.0+O	8	268.4	47.9
MF(49.55)	300.0-550.0+O	12	275.9	24.8
MF(50.05)	300.0-550.0	11	262.5	21.5

See Table 1 footnotes. O, origin used in fit.
† component does not trend towards origin.
‡ sample excluded in final polarity column.

Normal Polarity Characteristic Direction

All samples collected yielded a normal polarity component of magnetization with unblocking temperatures exceeding those of the inferred late Tertiary-recent components discussed above. After tectonic correction (strike=297.9°, dip=27.2°N), this component is similar to the Cretaceous normal polarity characteristic direction obtained throughout the Umbrian Apennines (Figure 17) (Tables 3-7). This component, however, is carried in different unblocking temperature ranges and by different minerals throughout the section. In interval I the component is isolated at temperatures between 150° and 450°C and is carried by magnetite. In interval II the component is carried between 150°C and an unblocking temperature ranging from 350° to 680°C, depending on

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TABLE 4. Bed A Reversed Characteristic Directions and Normal Overprints in Geographic Coordinates

		Normal Overprint				Reversed ChRM			
Sample	Meter	Temperature Range, °C	n	D, deg	I, deg	Temperature Range, °C	n	D, deg	I, deg
A5.0	5.00	300.0-425.0	6	251.2	6.6	600.0-650.0+O	5	152.4	-35.3
MF36.48	5.45	300.0-350.0	3	276.1	23.0	575.0-650.0+O	5	120.0	-38.4
A8.23	8.23	300.0-425.0	6	263.2	21.8	600.0-660.0+O	6	146.8	-35.4
A9.04	9.04	300.0-425.0	6	263.1	11.3	600.0-650.0+O	5	133.8	-55.8
A10.00	10.00	300.0-375.0	4	266.8	7.4	600.0-680.0+O	8	152.1	-46.1

See Table 1 footnotes. O, origin used in fit.

the particular sample; the dominant carrier appears to be hematite. Alternating field demagnetization experiments, however, suggest that magnetite also carries the normal polarity characteristic direction. In interval III, the component is mainly isolated between 300° and 580°C and is carried mainly by magnetite.

Reversed Polarity Characteristic Direction

Reversed polarity characteristic directions were found at 24 horizons throughout the section and were isolated (i.e., demagnetized) only after applying thermal demagnetization above 575°C (Figure 18, Table 6). We believe that all of these directions are carried by hematite which has a relatively discrete set of unblocking temperatures, generally between 625° and 690°C. Most intervals having reversed polarity characteristic magnetizations are found in interval II, but one horizon was found within interval I. In addition, interval III contains numerous horizons close to the boundary with interval II which revealed reversed polarity characteristic directions.

Statistical Treatment

The isolated directions were examined with the formal colatitude and longitude tests of *Fisher et al.* [1987] to determine whether the distributions were Fisherian [*Fisher*, 1953] (Table 11). These tests reveal that the hypothesis that the reversed polarity distribution (N=24) and the total normal polarity distribution (N=85) are Fisherian can not be rejected at the 95% confidence level. The means of these distributions, however, are not antipodal (Δ= 152°) (Tables 8 and 9) (Figure 18). The inverted inclinations are similar (δI = 4°), but the inverted declinations differ substantially (δD = 34°). One direction appears to be transitional between the two polarity groups as discussed further below.

When examined separately, the hypothesis that the normal polarity "overprints" on the samples having a reversed polarity characteristic remanent magnetization (N=24) are Fisherian also can not be rejected at the 95% confidence level (Tables 10 and 11), but the hypothesis can be rejected for the distribution derived from samples displaying only normal polarity (N=61). The latter fail the colatitude test but pass the longitude test, suggesting a departure from Fisherian behavior different from the cause of the nonantipodal character of the normal and reversed polarity directions.

An Unbiased Polarity Column

To examine the stratigraphic signature of the intervals carrying the reversed polarity ChRM, we have plotted inclination and declination of the highest unblocking temperature component isolated versus stratigraphic position for the Albian Contessa section (Figure 19; this sequence is derived from Tables 3 and 4). Caution is advised, however, in interpreting this plot (Figure 19) as a reliable sequence of ChRM polarity zones. Because a normal polarity magnetization is carried by each horizon measured, a potential bias exists in the recognition of intervals carrying reversed polarity ChRM. Specifically, a weak reversed polarity ChRM may be masked by a strong normal polarity component. Therefore, some of the changes in ChRM polarity between 30 and 40 m may be artifacts.

If a reversed polarity ChRM is being masked by a normal polarity component, upon demagnetization the normal polarity component should diverge from the origin. Samples from two zones between 30 and 40 m (32.75-35.36 m and 39.10-39.88 m) displayed such behavior. Unfortunately, samples from these zones exhibited directional instability at relatively low unblocking temperatures likely related to chemical changes in the rocks due to heating. It is difficult to determine in such samples whether a reversed polarity ChRM carried by hematite is present. Therefore, to limit any bias in our polarity column, we have marked these intervals as of uncertain polarity (see Table 3). We have excluded these samples to obtain our final polarity column (Figure 20; this sequence is derived from Figure 19 by deleting samples from 32.75-35.36 m and 39.10-39.88 m). If some of these uncertain intervals are actually of normal polarity, there are more polarity intervals than indicated in the final polarity column (Figure 20).

The polarity column shows two major intervals of reversed polarity along with several shorter intervals. Each reversed interval, with the exception of VC-1R (29.25 m), is defined

TABLE 5. Bed B Reversed Characteristic Directions and Normal Overprints in Geographic Coordinates

		Normal Overprint				Reversed ChRM			
Sample	Meter	Temperature Range, °C	n	D, deg	I, deg	Temperature Range, °C	n	D, deg	I, deg
B5.0	5.00	300.0-425.0	6	267.2	33.5	600.0-650.0+O	5	138.2	-40.8
MF35.86	5.48	300.0-475.0	8	251.2	15.9	600.0-625.0F	3	123.2	-56.3
B6.27	6.27	300.0-425.0	6	272.2	29.2	625.0-650.0F	3	148.8	-35.2
B6.75	6.75	300.0-425.0	6	279.2	27.3	600.0-660.0+O	6	131.0	-29.1
B7.77	7.77	300.0-425.0	6	265.6	15.1	600.0-680.0+O	8	134.4	-46.1
B12.30	12.30	300.0-425.0	6	270.6	12.9	625.0-680.0+O	7	129.5	-49.6

See Table 1 footnotes. O, origin used in fit; F, Fisherian average.

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TABLE 6. Reversed Characteristic Directions in Geographic Coordinates

Sample (m)	Temperature Range, °C	n	D, deg	I, deg
MF(29.35)	525.0-575.0+O	4	129.6	-67.9
MF(31.23)	600.0-650.0F	4	41.0	-43.8
MF(31.54)	625.0-680.0+O	6	121.1	-40.6
MF(31.60)	600.0-625.0+O	3	123.7	-46.9
MF(31.70)	600.0-670.0+O	7	125.1	-46.2
MF(33.45)	575.0-650.0+O	5	118.7	-54.7
MF(34.40)	615.0-650.0+O	5	126.1	-57.2
MF(34.55)	600.0-650.0+O	5	134.8	-47.6
MF(35.75)	600.0-640.0F	3	130.2	-43.3
BED B(35.86)†	600.0-680.0+O	6	134.9	-43.1
MF(36.05)	600.0-640.0+O	5	117.4	-20.7
MF(36.13)	600.0-640.0F	3	119.6	-34.1
BED A(36.48)‡	575.0-680.0+O	5	141.4	-42.9
MF(36.52)	600.0-670.0+O	7	152.5	-48.6
MF(37.05)	575.0-650.0	4	107.1	-23.8
MF(37.30)	600.0-655.0F	6	145.4	-36.5
MF(38.20)	600.0-660.0+O	7	114.7	-52.5
MF(38.42)	600.0-670.0	6	146.0	-53.8
MF(38.69)	600.0-670.0	6	135.9	-46.3
MF(38.94)	600.0-650.0	4	130.2	-48.5
MF(39.00)	600.0-670.0	6	103.2	-59.3
MF(39.62)	575.0-600.0+O	3	126.1	-24.5
MF(39.70)	600.0-640.0+O	4	97.8	-28.3
MF(40.03)	600.0-640.0	3	132.3	-33.3

See Table 1 footnotes. O, origin used in fit; F, Fisherian average.

† Average of specimens from bed B (Table 5).

‡ Average of specimens from bed A (Table 4).

by more than one sample. One sample (31.23 m) appears to record a transitional direction on the boundary of VC-2R. Faults in the section could have produced an unrecognized overlap in the section, and therefore one or more of the shorter intervals could be a sampling artifact. It is unlikely that the two longer intervals, VC-3R and VC-7R, are the duplication of a single zone.

TIMING OF MAGNETIZATION

The observed zones of reversed polarity characteristic magnetizations are laterally persistent, suggesting that they are recordings of reversed geomagnetic polarity intervals during deposition of the Contessa section. Since the reversed polarity magnetization may be carried solely in hematite, however, a remagnetization model may also explain the observed data. Recent work in North America has focused on the pervasive nature of Paleozoic remagnetizations now thought to be the consequence of fluid migration due to tectonic activity. Could the reversed polarity magnetizations observed in the Contessa section be the result of remagnetization? Below we assess these two hypotheses (primary magnetization and remagnetization) for the origin of the reversed polarity characteristic remanent magnetizations in the Albian Contessa section.

Model I: Primary Reversed Polarity Magnetizations

Channell et al. [1982] examined the relative timing of hematite and magnetite magnetizations in the Gubbio section, located 2 km SE of the Contessa section. Samples were collected close to previously determined polarity zone boundaries in Paleocene and Cretaceous Scaglia limestone. *Channell et al.* [1982] made two principal observations. First, the unblocking temperatures carrying a reversed polarity magnetization shifted toward higher unblocking temperatures as a reversal boundary was stratigraphically approached. Second, the highest unblocking temperature component carried in hematite recorded a field reversal at stratigraphic levels below the corresponding reversal recorded by magnetite. Based on these observations, *Channell et al.* [1982] developed a model for the acquisition of hematite magnetization in the Umbrian limestones. Hematite was interpreted as a diagenetic product growing from a goethite precursor. The nucleation of hematite was viewed as a continual process, penetrating down to some depth at which the local environmental conditions no longer permitted the formation of new hematite. In any collection of hematite grains, the largest grains would be the oldest. Since in single-domain hematite, grain size is directly related to unblocking temperature, the largest grains are also those grains having the highest unblocking temperatures. Hematite grains having very high unblocking temperatures acquire their magnetizations very early, earlier than the acquisition of a postdepositional remanent magnetization by magnetite. These very high unblocking temperature grains, therefore, are the best representation of a "primary" magnetization in the Umbrian rocks. Progressively smaller hematite grain with correspondingly lower unblocking temperatures carry a progressively later magnetization. Based on the data from the Scaglia limestones of the Gubbio section, hematite can nucleate down to a depth as great as 60 cm (after compaction) and therefore can carry a magnetization as much as 10^5 years older than the "primary" high-temperature hematite signal.

Following the model proposed by *Channell et al.* [1982] (we emphasize that here, and in the following discussions, we assume the grains are single domain) the high unblocking temperature hematite that carries the reversed polarity magnetizations in the Albian Contessa section could be the oldest and possibly the most reliable indicator of geomagnetic

TABLE 7. Normal Overprint Directions in Geographic Coordinates

Sample (m)	Temperature Range, °C	n	D, deg	I, deg
MF(29.35)	250.0-400.0	7	275.3	33.1
MF(31.23)	200.0-350.0	5	292.1	38.3
MF(31.54)	200.0-350.0	5	274.6	13.0
MF(31.60)	300.0-350.0	3	291.2	13.7
MF(31.70)	200.0-350.0	5	262.6	21.0
MF(33.45)	225.0-300.0	4	270.2	32.2
MF(34.40)	300.0-350.0	3	267.1	30.1
MF(34.55)	250.0-350.0	4	281.0	22.4
MF(35.75)	300.0-425.0	6	270.9	45.1
MF(35.86)B	300.0-475.0	6†	267.5	22.5
MF(36.05)	250.0-350.0	4	269.3	17.3
MF(36.13)	300.0-400.0	5	279.4	33.6
MF(36.48)A	300.0-350.0	5‡	263.9	14.1
MF(36.52)	300.0-350.0	3	253.3	22.7
MF(37.05)	300.0-500.0	9	276.1	23.9
MF(37.30)	250.0-350.0	4	288.3	18.9
MF(38.20)	250.0-375.0	5	282.4	26.4
MF(38.42)	250.0-375.0	5	281.8	25.3
MF(38.69)	250.0-350.0	4	287.8	26.8
MF(38.94)	250.0-400.0	6	274.3	17.4
MF(39.00)	300.0-450.0	7	271.3	34.2
MF(39.62)	300.0-425.0	6	273.1	28.8
MF(39.70)	325.0-475.0	7	272.0	33.3
MF(40.03)	300.0-475.0	8	282.6	30.6

See Table 1 footnotes.

† Fisherian average of specimens from bed B (Table 5).

‡ Fisherian average of specimens from bed A (Table 4).

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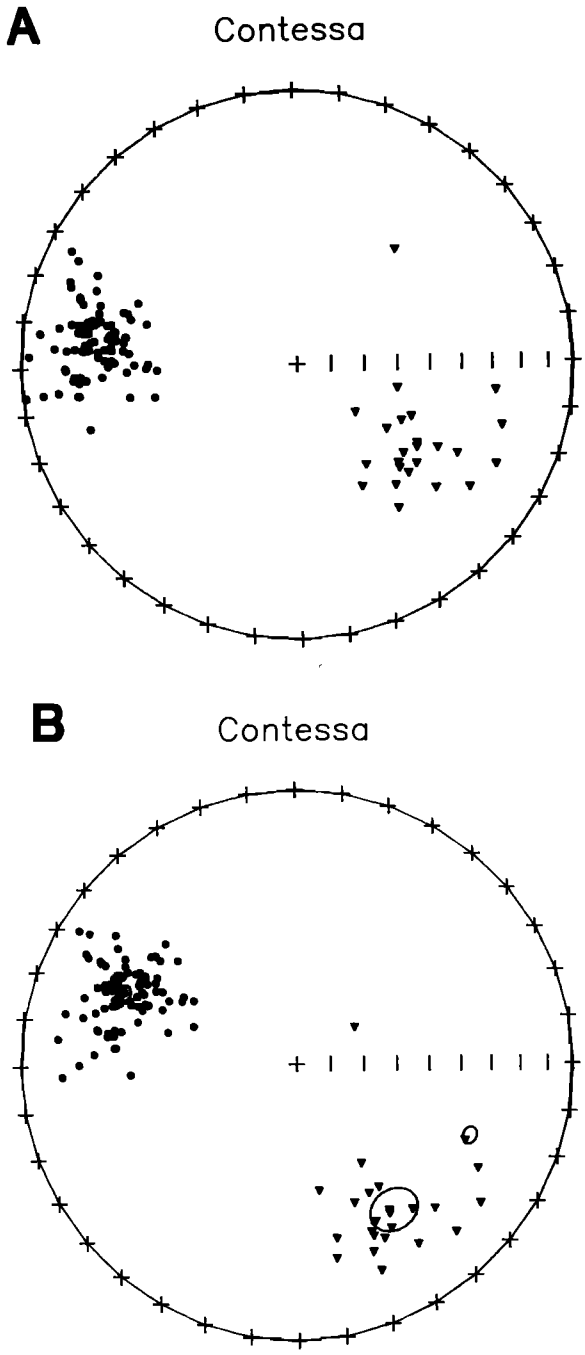


Fig. 18. Stereographic projection of normal- (Tables 3 and 7) and reversed-polarity (Table 6) directions from the Albian Contessa section. (*a*) Geographic (in situ) coordinates. (*b*) Stratigraphic coordinates (after tilt correction). Solid circles represent 95% confidence intervals. Also shown is the inverted normal polarity confidence interval for comparison with the confidence interval for the reversed polarity directions.

field polarity during initial deposition. Importantly, this high unblocking temperature hematite carries both normal and reversed polarity magnetizations.

Unlike the data from the Scaglia limestone of the Gubbio section [*Channell et al.*, 1982], our data from the Albian Contessa section do not exhibit a clear shift toward higher unblocking temperatures as a reversal boundary is approached. Instead, complex directional behavior during progressive

TABLE 8. Averaged Directions for Geographic Coordinates

Average	n	I, deg	D, deg	k	α_{95}
1 Brunhes overprint	8	66.4	358.6	81.9	5.5
2 Matuyama overprint	8	-53.3	199.2	32.7	8.7
3 Normal ChRM	61	26.9	277.8	40.5	2.8
4 Normal overprint	24	26.3	275.3	46.8	4.2
5 Normal total (3+4)	85	26.8	277.0	41.3	2.4
6 Reversed ChRM	24	-45.4	123.7	18.3	6.7

See Table 1 footnotes.

thermal demagnetization between 500° and 575°C (interval II) is observed, suggesting the removal of multiple (normal and reversed polarity?) components (Figures 10, 12, and 13).

There are reasons, however, to expect differences between the Gubbio Scaglia limestone and the Marne a Fucoidi. The high unblocking temperature hematite magnetizations in the Contessa section coincide roughly, although not exactly, with two reddish bands in the outcrop (Figure 21). These reddish marls can be correlated directly to other sections in Umbria such as that recovered by drilling near Piobicco [*Premoli Silva et al.*, 1983; *Pratt and King*, 1986; *Herbert et al.*, 1986; *Erba*, 1988]. The reddish color is likely a pigmentary hematite, having small grain sizes and reflected in our magnetic measures as hematite with low unblocking temperature ranges (i.e., that hematite in the coercivity fraction magnetized at 0.15 T in interval II). This pigmentary hematite may reflect the oxidation state of the seafloor at the time of deposition. Interbedded within the reddish marls are occasional black shales [*Tornaghi et al.*, 1989], indicating a rapid shift between highly oxidizing and highly reducing conditions. In contrast, the Scaglia limestone studied by *Channell et al.* [1982] most likely records deposition under more uniform oxidation conditions.

We hypothesize that the reversed ChRM at Contessa may be carried by hematite acquired during two intense oxidation events as evidenced by the two reddish marl units. Magnetite and fine-grained hematite may record magnetizations acquired slightly after deposition (i.e., normal polarity overprints on samples with reversed ChRM). This hypothesis also implies that the hematite could have grown down to a depth of at least 60 cm [*Channell et al.*, 1982] during the later oxidation event, possibly overprinting a portion of interval II (Figures 19 and 21). We note that many greenish-grey beds carry a reversed polarity magnetization carried by hematite.

The only other reversed polarity zone identified in Umbria strata deposited during the K-N Superchron is the ISEA interval of late Aptian age [*VandenBerg et al.*, 1978]. This interval is also associated with reddish marls, the "couche rouge", and has a magnetization dominated by hematite as denoted by IRM acquisition data [*Lowrie et al.*, 1980*a*]. The primary magnetization model outlined above fails to account for this coincidence, other than by suggesting that hematite bearing zones are more reliable recorders of short polarity intervals than are highly reduced pelagic facies (i.e.,

TABLE 9. Averaged Directions for Stratigraphic Coordinates

Average	n	I, deg	D, deg	k	α_{95}
3 Normal ChRM	61	32.5	293.3	40.5	2.8
4 Normal overprint	24	33.1	290.7	46.9	4.2
5 Normal total	85	32.6	292.6	42.3	2.3
6 Reversed ChRM	24	-36.6	146.7	18.3	6.7

See Table 1 footnotes.

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TABLE 10. Lateral Consistency Test, Geographic Coordinates

Average	n	M_U	Pass?	M_E	Pass?	I, deg	D, deg	k	α_{95}
Bed A normal overprint	5	0.883	yes	0.566	yes	14.1	263.9	48.3	9.0
Bed A reversed ChRM	5	0.867	yes	0.728	yes	-42.9	141.4	35.4	10.5
Bed B normal overprint	6	0.815	yes	0.838	yes	22.5	267.5	43.5	8.7
Bed B reversed ChRM	6	1.122	yes	0.642	yes	-43.1	134.9	47.8	8.3

See Table 1 footnotes. M_U, M_E after *Fisher et al.* [1987]. Yes, hypothesis that distribution is Fisherian cannot be rejected at 95% confidence.

magnetite may be produced as an authigenetic mineral in reduced facies and hence record a later magnetization). There is not a one-to-one correlation between hematite and polarity in the Albian Contessa section (nor, apparently, in the couche rouge); hematite carries both normal and reversed polarity magnetizations.

Model II: Remagnetized Reversed Polarity Magnetizations

The possibility that hematite is the sole carrier of reversed polarity characteristic magnetizations suggests that models for the precipitation and magnetization of hematite long after deposition could also explain these data from the Valle della Contessa. An analogy for such a remagnetization scenario might be found in the numerous studies of North America Paleozoic carbonates which document overprints acquired up to 150 m.y. after deposition during the Permo-Carboniferous Reversed Polarity Superchron (PC-R or Kiaman). While the magnetization in many of these carbonates appears to be carried in two distinct grain size fractions of magnetite [*Jackson*, 1989], examples of remagnetization directions carried by hematite have also been noted (Table 12).

In many of these studies where hematite carries the remagnetization, textural relations have provided strong supporting evidence for the creation of a CRM by precipitation of hematite during fluid flow. For example, *Elmore et al.* [1985] found a Kiaman remagnetization carried by hematite which they suggest precipitated due to infiltration of oxidizing fluids that caused dedolomitization. In a review of Kiaman remagnetizations, *McCabe and Elmore* [1989] concluded that the fluids related to hematite precipitation could have been basinal or meteoric and that when present, remagnetizations carried by hematite are not necessarily pervasive. Remagnetized rocks are sometimes found proximal to nonremagnetized rocks with differences perhaps related to the state of cementation of the rock [*McCabe and Elmore*, 1989]. Rather than being pervasive, the secondary hematite appears to be associated with specific pathways for fluid flow and specific diagenetic events.

How applicable are the studies in Paleozoic carbonates to our new paleomagnetic results from the Albian Contessa section? First, we should note that while the Paleozoic examples discussed are carbonates, there are great differences between these rocks and the Umbrian pelagic limestone sequence. The Paleozoic examples are from shallow marine facies (e.g., subtidal, intertidal, and supratidal limestone and dolomite), while the rocks from the Umbrian sequence that have yielded reliable magnetostratigraphic results are well-cemented pelagic limestones. The general difference between the two rock types is notable since conduits for fluid flow might be more restricted in the micritic Umbrian rocks.

While there do seem to be important differences in gross lithology, the mechanism of remagnetization carried by hematite in the North American Paleozoic carbonates may be applicable to the Contessa section. Based on the analogy to Kiaman remagnetizations, we can develop the following scenario. Oxidizing fluids flowed through the Marne a Fucoidi at Contessa during a reversed polarity chron, at least 18 m.y. after deposition, causing the precipitation of hematite which acquired a reversed polarity magnetization. Fluid flow affected only a portion of the measured section and occurred along specific beds while the section was flat-lying. If acquired during reversed polarity chron 33R, the remagnetization would have occurred at a subbottom depth of at least 200 m, based on the compacted overburden calculated from the Moria section [*Alvarez and Lowrie*, 1978].

The remagnetization scenario is appealing since it seemingly explains (1) the correspondence between hematite and reversed polarity magnetizations and (2) the nonantipodal characteristics of the normal and reversed polarity ChRM since the two polarities could be acquired in different time intervals. In fact, the reversed polarity ChRM data agree more closely with data from polarity chrons younger than the

TABLE 11. Tests for Fisheran Distribution

Average	n	M_U	Pass?	M_E	Pass?
Brunhes	8	1.307	no	1.000	yes
Matuyama	8	0.840	yes	0.847	yes
Normal ChRM geographic	61	1.056	yes	1.425	no
Normal overprint geographic	24	0.779	yes	0.730	yes
Normal total geographic	85	1.095	yes	1.197	no
Reversed ChRM geographic	24	1.183	yes	0.915	yes
Normal ChRM stratigraphic	61	1.052	yes	1.413	no
Normal overprint stratigraphic	24	0.777	yes	0.722	yes
Normal total stratigraphic	85	1.070	yes	1.005	yes
reversed ChRM stratigraphic	24	1.191	yes	0.913	yes

See Table 1 footnotes. M_U, M_E after *Fisher et al.* [1987]. Yes, hypothesis that distribution is Fisherian cannot be rejected at 95% confidence. No, hypothesis that distribution is Fisherian can be rejected at 95% confidence.

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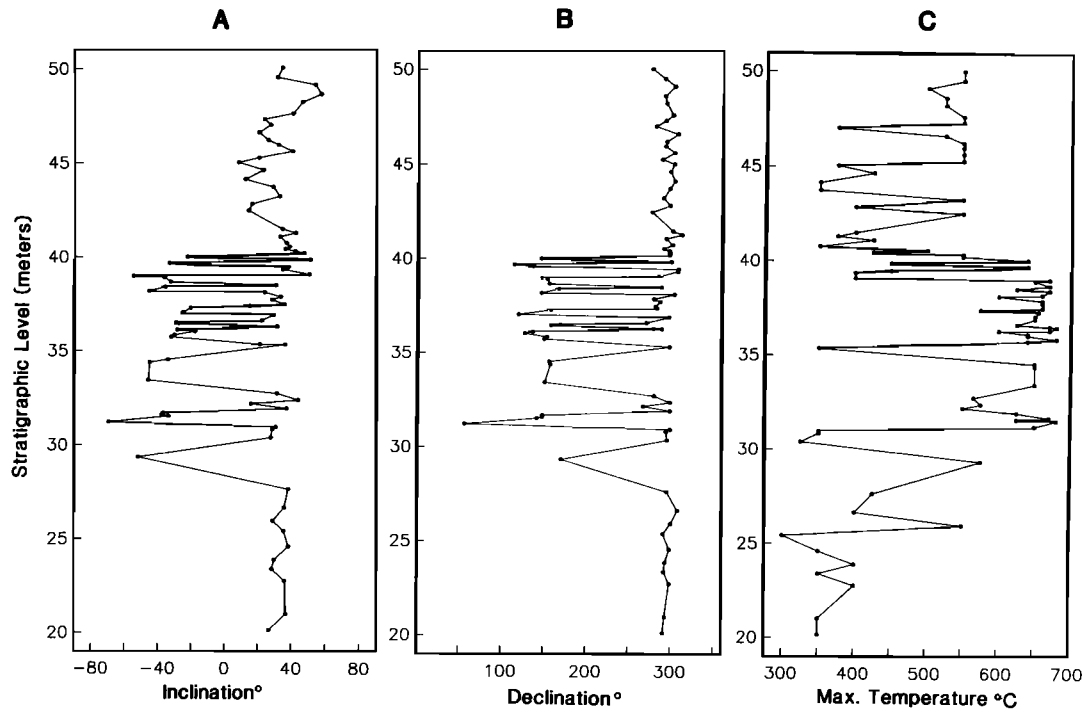


Fig. 19. Normal- and reversed-polarity (*a*) inclinations and (*b*) declinations versus meter level in the Albian Contessa section. (*c*) The maximum unblocking temperature used to obtain the data (Figures 19*a* and 19*b*).

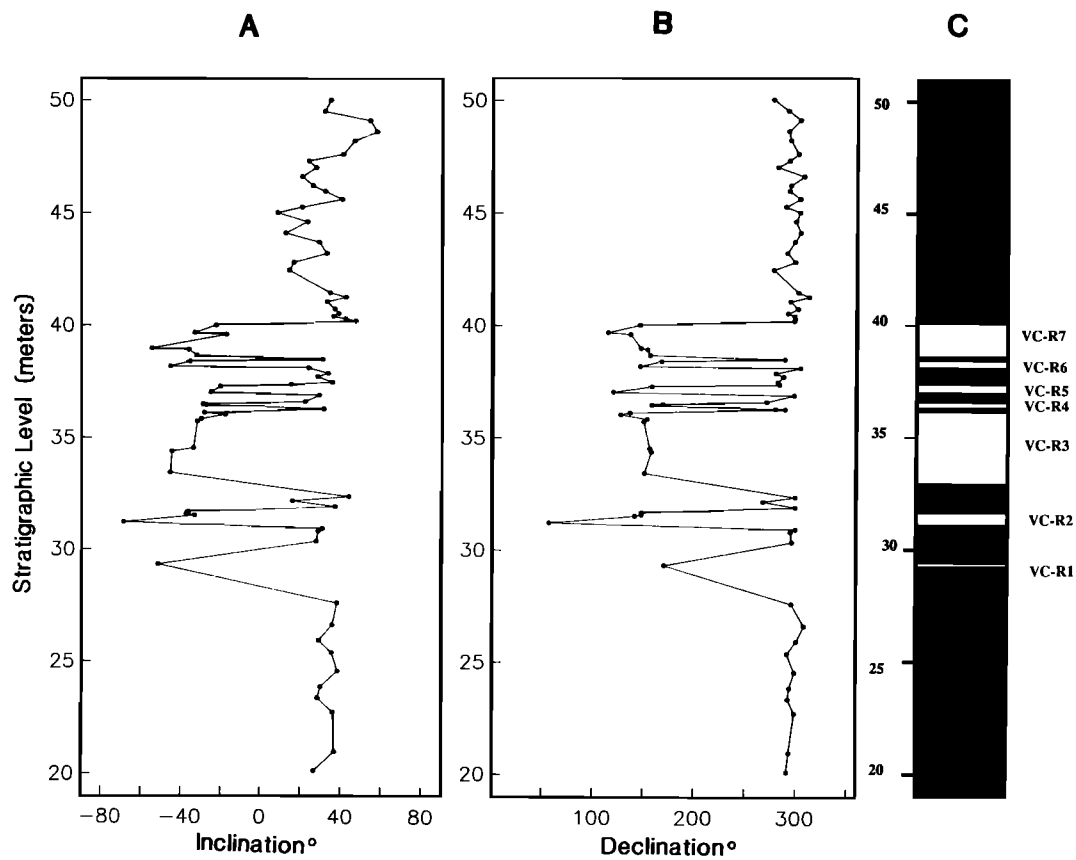


Fig. 20. Unbiased polarity column for the Albian Contessa section obtained by omitting samples from rock magnetic intervals II and III having undefined directions at high unblocking temperatures (see text and Table 3). Stippled region is of uncertain polarity due to lack of samples.

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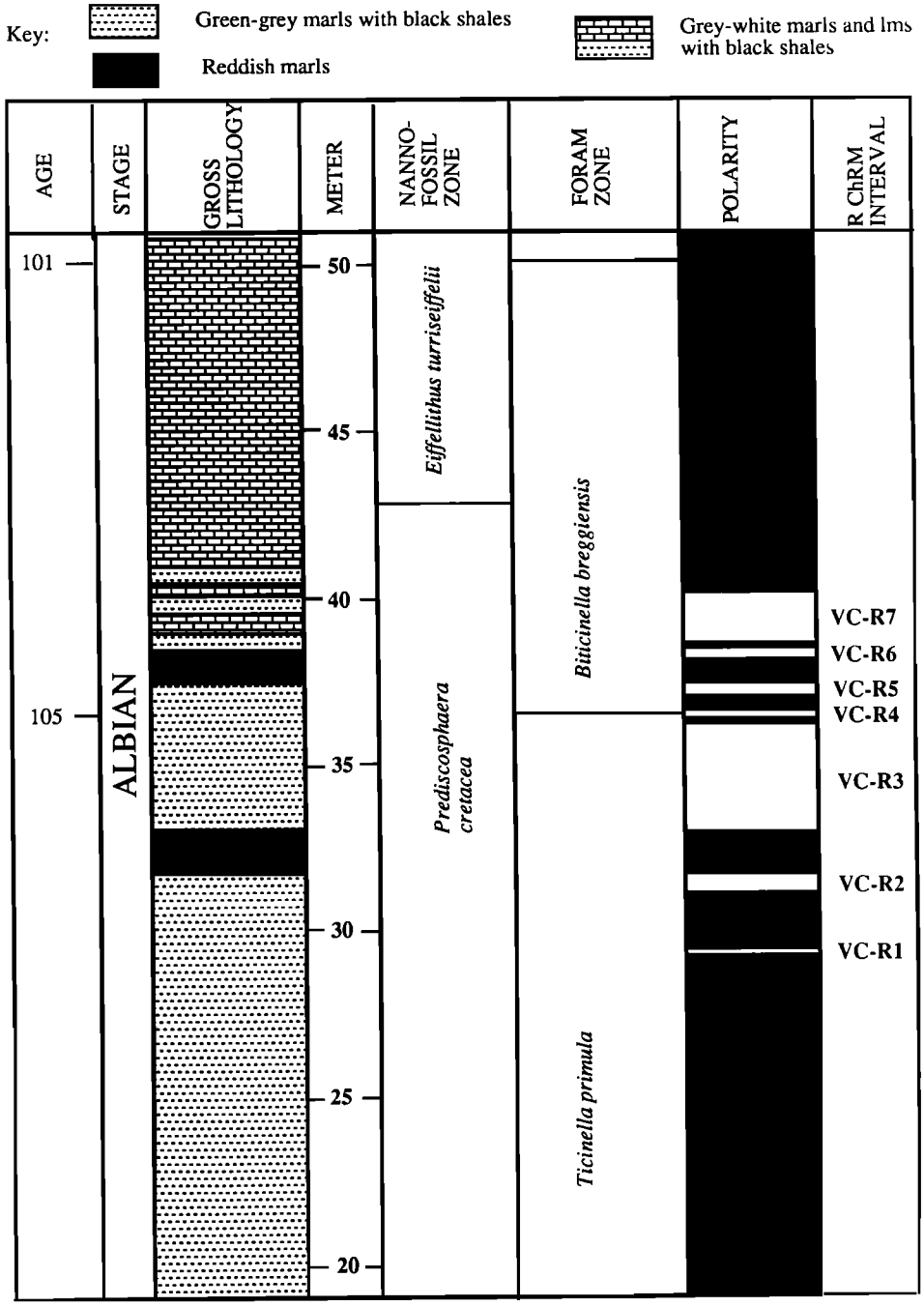


Fig. 21. Interpreted polarity versus stage, gross lithology, and age. Absolute ages after *Sliter* [1991].

TABLE 12. Kiaman Remagnetization Carried by Hematite in Paleozoic Carbonates

Unit	Age	Unblocking Range °C	Reference
St. George Group.	O	300-530	*Deutsch and Prasad* [1987], Figure 6
Taum Sauk Limestone	C	285-500	*Dunn and Elmore* [1985] Figure 7
Kindblade Formation†	O	194-609	*Elmore et al.* [1985] Figure 3
Peerless Fm.	C	<548->670	*Peck et al.* [1986] Figure 3

† Also contains a component of unkown orgin at unblocking temperatures above 609°C; age abbreviations: C, Cambrian; O, Ordovician.

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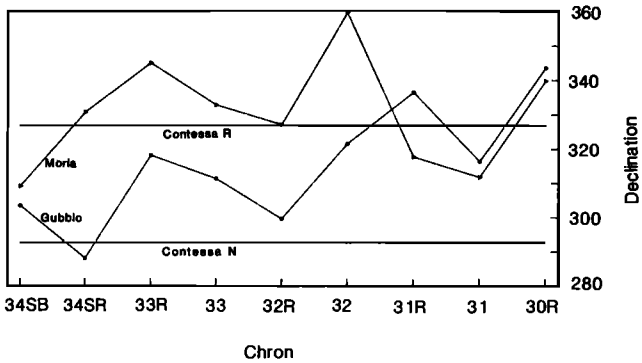


Fig. 22. Comparison of declination values from Moria (solid triangles), Gubbio (solid circles) [*Alvarez and Lowrie*, 1978] and the Albian Contessa section (solid lines, this work) shown versus polarity chron (34SB, average from the Scaglia Bianca of the K-N Superchron; 34SR, average from the Scaglia Rossa, K-N Superchron). Data from adjacent polarity zones are not antipodal. Normal polarity mean direction from the Contessa section is close to the K-N Superchron average from Gubbio, while reversed polarity mean agrees better with averages from younger chrons.

K-N Superchron as defined in the Gubbio section (Figure 22).

When examined critically, however, the model has several deficiencies. The model fails to explain the discrete unblocking range of hematite carrying the reversed polarity magnetizations. The results from Contessa differ from studies of Paleozoic rocks in that the reversed polarity ChRM is carried by very high unblocking temperature hematite. In most examples of remagnetization, a greater unblocking temperature range is affected (Table 12). The model also fails to explain the occurrence of both normal and reversed polarity magnetizations carried by the very high unblocking temperature hematite other than by coincidence: some intervals must carry a primary direction of only normal polarity at very-high unblocking temperatures, while others carry a remagnetization of only reversed polarity at the same very high unblocking temperatures.

One way to explain the normal magnetizations at very high unblocking temperatures is to suggest that the remagnetization occurred over sufficient time to encompass a polarity change. When this refinement is added, however, the remagnetization model can not account for the nonantipodal character of the normal and reversed polarity directions as outlined above. For example, during the growth of the secondary hematite, the field is of reversed polarity. Hematite growth continues so that the largest grains are the oldest and carry a reversed polarity magnetization. The remagnetization occurs long after initial deposition so the direction of the inverted reversed ChRM differs from the primary normal polarity magnetization (acquired during the K-N Superchron). At some point the (remagnetization) field reverts to normal polarity. In certain beds the hematite growth starts only after the polarity change but must proceed long enough so that very large grains having very high unblocking temperatures are precipitated. But hematite growth should be a continual process with small hematite grains constantly forming. Futhermore, from our rock magnetic data we can infer that a significant amount of hematite having smaller grain sizes is present. Yet none of the smaller hematite grains (having lower unblocking temperatures) has a magnetization predicted by this scenario: a normal polarity magnetization with a declination like that exhibited by the inverted reversed polarity ChRM data set. If a remagnetization occurred long after deposition when the field had a different direction and proceeded long enough to include a change in polarity, the normal polarity data should be bimodal in declination. Instead it is azimuthally uniform.

DISCUSSION

Both hypotheses (primary magnetization and remagnetization) fail to account completely for the properties of the reversed polarity characteristic magnetizations. The principal deficiency of both models is in explaining the magnetization of only the largest hematite grains. In light of these discrepancies, several areas merit further discussion. If the zones of reversed polarity ChRM are primary magnetizations and are of sufficient duration (>100 kyr), corresponding zones should be found on a worldwide basis. Below we review the duration of the potential reversed polarity intervals and their correlation to previous studies. Assignment of the reversed polarity characteristic magnetizations to a post-K-N Superchron magnetization relies on the accuracy of the isolated Contessa directions and the reference directions. Below we also review the reliability of the differences between the mean normal and reversed polarity directions and Umbrian reference directions.

Duration of Potential Reversed Polarity Interval

If the zones of reversed polarity ChRM in the Albian Contessa section are recordings of reversed polarity intervals, we would like to obtain some estimate of the duration of these intervals. The reversed polarity ChRM intervals can be tied to geological stage and absolute age using the micropaleontological data (Figure 21). The only complete biozone represented is the *Biticinella breggiensis* foraminiferal zone. The first appearance datum (FAD) of *B. breggiensis* (36.48 m) has been estimated to occur at 105 Ma [*Sliter*, 1991], while the FAD of *Rotalipora ticinensis* (50.05 m), which identifies the *R. ticinensis* Zone, has been estimated at 101 Ma. These ages rely on the absolute age calibration of biozones given in the *Harland et al.* [1982] time scale. The revised time scale of *Harland et al.* [1990] incorporates only minor changes in the relevant portion of the Albian. The ages discussed above suggest a sedimentation rate of approximately 3.4 m/m.y. for the Albian Contessa section. These sedimentation rates suggest durations of the polarity intervals recorded by the reversed polarity ChRM which range from 973 to 59 kyr (Table 13).

A second method can be employed to estimate the duration of the Contessa reversed polarity intervals. Cyclic sedimentation has long been recognized in the Valle della Contessa [*Schwarzacher and Fischer*, 1982, Figure 7]. Analyses in the Scaglia Bianca and Maiolica Limestones have recognized groups of four to six beds, termed "bundles", which are thought to coincide with the Earth's 100-kyr eccentricity cycle [*Schwarzacher and Fischer*, 1982; *Fischer and Schwarzacher*, 1984; *Arthur et al.*, 1984; *de Boer and Wonders*, 1984]. These bundles can also be recognized in the Marne a Fucoidi (Figure 23).

The most straightforward estimate of duration of the reversed polarity intervals recorded in the Contessa section would be the number of bundles per polarity zone. Unfortunately, an accurate count was not possible throughout the entire section due to small-scale faulting and difficulties in

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TABLE 13. Duration of Potential Reversed Polarity Intervals

Event	Range, m	Thickness, m	Duration, kyr	
			Estimate 1	Estimate 2
VC-1R	29.35	indeterminate	indeterminate	indeterminate
VC-2R	31.1-31.8	.7	206	175
VC-3R	32.9-36.2	3.3	973	825
VC-4R	36.4-36.6	.2	59	50
VC-5R	37.0-37.35	.35	103	88
VC-6R	38.17-38.47	.3	88	75
VC-7R	38.61-40.10	1.49	439	373

Estimate 1 is based on paleontological data after *Sliter* [1991]. Estimate 2 uses Milankovitch-like bedding cyclicity.

access. Instead, we rely on a less satisfactory approach; estimating a mean sedimentation rate. Slightly over five bundles can be identified per 2-m interval (Figure 23), suggesting a sedimentation rate of approximately 4 m/m.y. This rate suggests durations of the Contessa reversed intervals ranging from 825 to 50 kyr (Table 13).

We urge caution in the application of these durations for several reasons. There are substantial uncertanties in the absolute ages used for the paleontological data. In addition, minor faults are present and changes in sedimentation are likely within the section. Nevertheless, it appears that some of the zones represent time intervals greater than 100 kyr. If these reversed polarity zones are primary recorders of reversed geomagnetic field polarity, they would represent subchrons (as opposed to short polarity events or "tiny wiggles") not previously recognized within the K-N Superchron.

Correlation of Potential Reversed Polarity Intervals

Reversed polarity magnetizations have been reported previously from a number of Albian sequences. These data fall into two categories: those having both inclination and declination control and those based on inclination-only data. The former include most field studies, while the latter include most studies of cores recovered by the DSDP and its successor, the Ocean Drilling Program. Reports of reversed polarity magnetizations from DSDP Albian sediments are reviewed in terms of their reliability and possible correlation with intervals of reversed polarity ChRM from the Albian Contessa sequence in the appendix (Table 14).

In summary, the presence of an Albian mixed polarity interval based solely on DSDP examples is tenuous and explains why such an interval has not been adopted in recent studies [e.g., *Harland et al.*, 1982; *Kent and Gradstein*, 1986; *Harland et al.*, 1990]. All examples could be artifacts due to accidentally inverted core pieces, core sections, or entire cores. An Albian mixed polarity interval has previously been suggested by *van Hinte* [1976] using data from DSDP site 263 in the Indian Ocean [*Green and Brecher*, 1974; *Jarrard*, 1974]. If real, the "site 263" mixed polarity interval occurs at a slightly later time than those recorded at the Contessa section and records reversed polarity intervals of shorter duration. Possible field reversals recorded at DSDP site 400 in the North Atlantic fall close to the Contessa events (Figure



Fig. 23. Detail of bedding in the Albian Contessa section. Groups of four to six beds, called "bundles", thought to reflect the Earth's 100-kyr eccentricity cycle [*Schwarzacher and Fischer*, 1982], are labeled (black arrows). Slightly over five bundles are identified per 2-m interval (ruler shown).

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TABLE 14. Possible Brief Reversed Polarity Intervals: K-N Superchron

Event	Class	Stage	Age, Ma	Location	Reference
ISEA	A	Aptian	115	Valdorbia, Italy	*Lowrie et al.* [1980*a*]
463	A	Aptian	115	DSDP site 463	*Tarduno* [1990]
402-2R	B	Aptian	115	DSDP site 402	*Tarduno* [1990]
317	B	Aptian	115	DSDP site 317	*Tarduno* [1990]
400-5R	B	Aptian	115?	DSDP site 400	*Hailwood* [1979]
402-1R	B	Aptian-Albian	113?	DSDP site 402	*Tarduno* [1990]
400-1R	B	Albian	98-104?	DSDP site 400	*Hailwood* [1979]
400-2R	B	Albian	98-104?	DSDP site 400	*Hailwood* [1979]
400-3R	B	Albian	104-107?	DSDP site 400	*Hailwood* [1979]
400-4R	B	Albian	104-107?	DSDP site 400	*Hailwood* [1979]
260-1R	B	Albian	104-107?	DSDP site 260	*Jarrard* [1974]
263-1R	B	Albian	98-104?	DSDP site 263	*Jarrard* [1974]
263-2R	B	Albian	98-104?	DSDP site 263	*Jarrard* [1974]
263-3R	B	Albian	98-104?	DSDP site 263	*Jarrard* [1974]
263-4R	B	Albian	104-107?	DSDP site 263	*Jarrard* [1974]
ZC	A	Albian	98-104?	Zopilote Canyon	*Urrutia-Fucugauchi* [1988]
VC-1R	A	Albian	107.0	Contessa, Italy	this work
VC-2R	A	Albian	106.5	Contessa, Italy	this work
VC-3R	A	Albian	105.6	Contessa, Italy	this work
VC-4R	A	Albian	105.0	Contessa, Italy	this work
VC-5R	A	Albian	104.8	Contessa, Italy	this work
VC-6R	A	Albian	104.5	Contessa, Italy	this work
VC-7R	A	Albian	104.2	Contessa, Italy	this work

24). Nevertheless, few deep-sea sections have recovered the biozones represented at Contessa. At present, we are aware of no paleomagnetic study (reporting mixed or constant normal polarity magnetizations) of a DSDP or ODP site (or on-land section) that has been analyzed in sufficient detail to serve as a rigorous test by correlation of the primary nature of the Contessa reversed polarity intervals.

Nonantipodal Distributions

The evaluation of whether reversed and normal polarity magnetizations are antipodal is commonly taken as a consistency test for the reliability of paleomagnetic directions [e.g., *Gordon et al.*, 1984]. If such data are not antipodal, it is concluded that an unremoved overprint is present. With both Brunhes and Matuyama field overprints identified in the Contessa section, often within a single specimen, it is reasonable to question the accuracy of our representation of the Cretaeous geomagnetic field direction by the thermal demagnetization data. An argument against the presence of unremoved overprints is the lack of noticeable streaking in the normal and reversed data sets. Nevertheless, curvature of the demagnetization paths are sometimes observed. This is especially apparent in the declination values used to define the normal polarity component (Figures 9-14). It is possible that thermal demagnetization has not completely removed the effects of the "Matuyama" field overprint.

The question of how well Cretaceous data conform to a model of antipodal reversals was addressed by *Alvarez and Lowrie* [1978] in their study of the Moria section. Most of the normal polarity intervals studied were found not to be antipodal to adjacent reversed polarity intervals (Figure 22). Demagnetization treatments were not as rigourous as those applied in this new work (i.e., only AF demagnetization was applied) which may explain why the Moria data are not Fisherian [*Alvarez and Lowrie*, 1978]. In addition, complications may be present due to likely tectonic rotations during deposition [*Lowrie and Alvarez*, 1975]. There do seem to be trends in the declination data seen both in the Moria and Gubbio section [*Alvarez and Lowrie*, 1978].

From our data and a review of the previous investigations it is not possible to definatively ascribe the nonantipodal nature of the normal and reversed polarity directions to any single cause. Unfortunately, both the Contessa directions and the reference directions (i.e., Gubbio and Moria) may be biased by unremoved overprints. At present, however, it is reasonable to assume that the differences between the normal and reversed polarity ChRM directions at Contessa are exaggerated by unseparated components such as the "Matuyama" field overprint.

SUMMARY AND CONCLUSIONS

The two models (primary magnetization and remagnetization) proposed to explain the intervals of reversed polarity ChRM isolated in the Albian Contessa section can be assessed by the following observations: (1) the reversed polarity magnetizations are isolated (i.e., demagnetized) only at very high unblocking temperatures; (2) very high unblocking temperature hematite also carries normal polarity magnetizations; (3) the reversed polarity magnetizations are laterally continuous and the normal and reversed polarity magnetizations form stratigraphic zones; (4) the normal and reversed polarity data sets are not antipodal ($\Delta = 152^o$); and (5) the reversed polarity ChRM (large grained hematite) corresponds approximately with two stratigraphic bands of reddish marls (pigmentary hematite).

If the reversed polarity ChRM is a remagnetization, the process would have to be one that could affect only a portion of the stratigraphic section, along specific beds before tilting. By analogy with Kiaman remagnetizations, fluids could have flowed through the section long after deposition resulting in the growth of hematite that acquired a reversed polarity magnetization only in those beds which acted as pathways for the fluids. A candidate for the time of remagnetization is reversed polarity chron 33R (83-79 Ma). In this scenario, the reversed polarity ChRM is not antipodal to the normal polarity direction because it was acquired at a much later time.

Scrutiny reveals several flaws in this remagnetization

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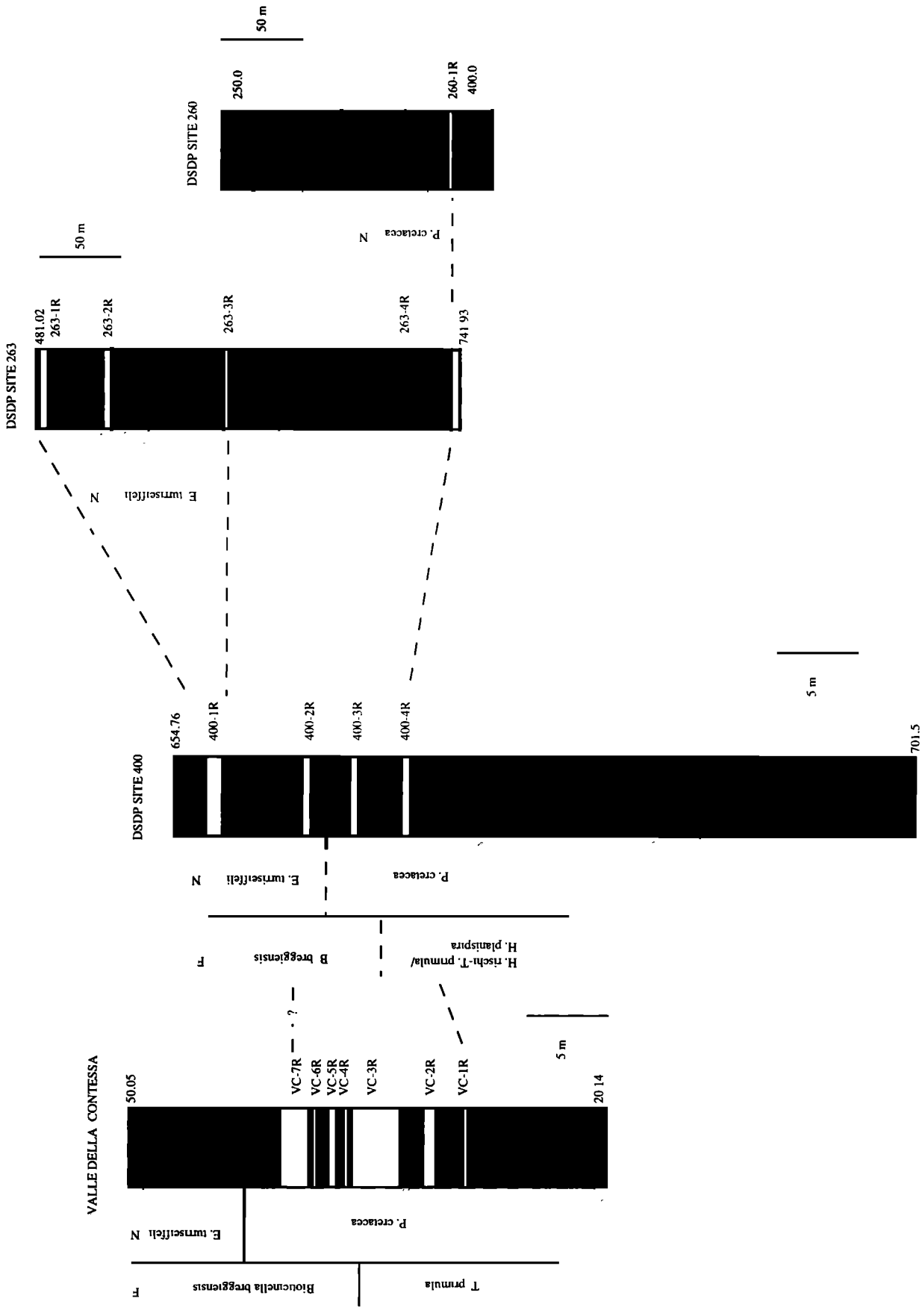


Fig 24. Tentative correlation of the Contessa polarity sequence with DSDP sites 400 and site 260. Stippled, no core recovery. Data from site 263, if valid, record geomagnetic reversals younger than those recorded at Contessa. N, nannofossil date; F, foraminiferal date.

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model. If secondary hematite grew as described, we should expect a range of grain sizes, with the largest having grown the longest, ranging down to very small grains which just started to nucleate when fluid flow ceased. The reversed polarity magnetizations, however, are isolated only at very high unblocking temperatures, above 600°C, indicating that these magnetizations are carried only by the largest hematite grains. The remagnetization model also fails to explain the presence of high unblocking temperature hematite carrying normal polarity magnetizations. If a remagnetization, the mechanism must be one previously unappreciated but nevertheless of great importance since it can produce magnetization patterns which resemble reliable recordings of polarity intervals and can affect only a restricted range of hematite grain sizes.

Alternatively, the reversed polarity ChRM may have been acquired during intense seafloor oxidation episodes in the Albian (107-104 Ma) evidenced by the reddish sediments and may record unrecognized brief geomagnetic reversed polarity intervals. This model is consistent with the largest hematite grains carrying the most accurate measure of field polarity since they are the first to nucleate, as has been observed in detailed studies near polarity transitions [*Channell et al.*, 1982]. The nonantipodal character of the reversed and normal polarity directions may be due to unremoved components, especially a "Matuyama" field overprint. If a primary magnetization, the reversals recorded in the Contessa section imply a substantial revision of the mid-Cretaceous geomagnetic chronology (Figure 25). Some of the reversed intervals appear to be of short duration (100 kyr or less) which would account for their lack of resolution in marine magnetic anomaly records. Two intervals are longer (>100 kyr), however, and should be identifiable in both detailed stratigraphic section and marine magnetic anomaly surveys. A candidate in the seafloor record is a marine magnetic anomaly labeled 35R identified by *Ladd* [1974] in the South Atlantic.

An obvious deficiency of the primary magnetization model is the apparent correlation of reversed polarity ChRM with hematite. Therefore, a remagnetization scenario can not be excluded until the reversals identified at Contessa are identified elsewhere. The primary magnetization model predicts that several intervals of reversed polarity magnetization should be found near the boundary of the *Biticinella breggiensis* and *Ticinella primula* foraminiferal zones, within the *Prediscosphaera cretacea* nannofossil zone of mid-Albian (107-104 Ma) age.

Note added in proof. Recently a paper by *Hambach and Krumsiek* [1989] has become available describing a series of reversed polarity intervals of Albian and Cenomanian age from the study of drill cores in the Ruhr area (western Germany). Unlike the stratigraphic range studied at Contessa (middle Albian), the Cenomanian has been sampled in detail in several sections from the Umbrian Apennines and Southern Alps [e.g. *Alvarez and Lowrie*, 1978; *Channell et al.*, 1979*a*]. The report of *Hambach and Krumsiek* [1989] is surprising since in these previous studies no intervals of reversed polarity have been identified in Cenomanian-age sediments. The only example of Cenomanian reversed polarity intervals comes from the Quero-Schievenin section [*Vandenberg and Wonders*, 1980], but this section may have complications due to slumping (H. Thierstein, personal communication, 1989). Interestingly, *Hambach and Krumsiek* [1989] describe the mineral carrier in their study as having a high coercivity

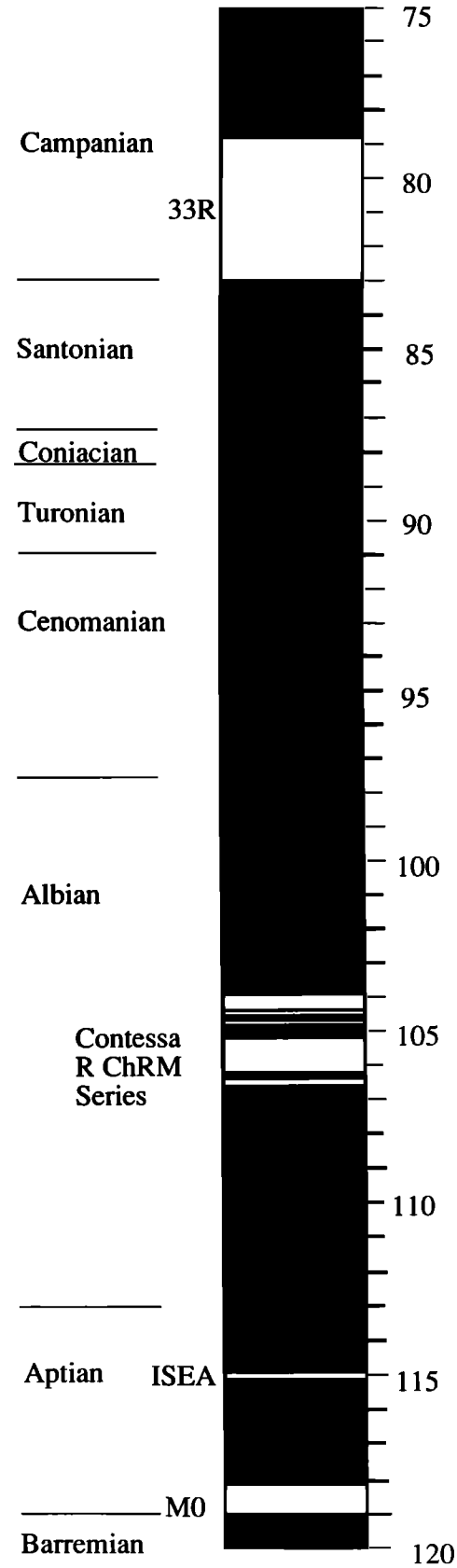


Fig. 25. Possible Mid-Cretaceous Polarity Time Scale. Absolute ages after *Harland et al.* [1982]. ISEA after *Tarduno* [1990]. Contessa reversed ChRM series, this work.

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and low unblocking temperature. Such low unblocking temperatures in the Albian Contessa section were found to record an alternation of normal "Brunhes" field overprints and reversed "Matuyama" field overprints. Based on the lack of substantial reversed polarity intervals in other Cenomanian sections and by analogy with the overprints observed in the Contessa section, we suggest that the reversed polarity intervals observed from the Ruhr area are artifacts of overprinting in a Matuyama (or late Tertiary) field. We invite further studies to clarify this issue.

APPENDIX

The magnetostratigraphic study of Cretaceous DSDP and ODP sections encounters several disadvantages when compared to the study of their on-land counterparts. Full recovery is rare for indurated Cretaceous deep-sea sediments. In recording depths during drilling, it is convention to group the unrecovered depth of a core at the base of the recovered portion. The "unrecovered portion", however, could have accumulated at many gaps within the total depth penetrated. Therefore, a misleading picture of the relative widths of paleontological and polarity zones can result when recovery is not complete. This problem can be acute when brief polarity events represented by one meter or less are the focus of study. In addition, most Cretaceous DSDP and ODP sections obtained by rotary drilling are broken into pieces during drilling. Accidentally inverted pieces produce the false impression that reversals have occurred. In rare instances, entire DSDP sections have been accidentally inverted [see *Tarduno*, 1990]. Occasionally, an overprint is present on DSDP cores, such as that related to the present-day field. Demagnetization of such an overprint can be used to define a proxy for declination (Δ^o of *Tarduno et al.* [1989]) to distinguish actual reversed magnetizations from the above artifacts. If such an overprint is not present, it is not possible to make this distinction.

Some deep-sea sections have certain advantages over on-land sections such as higher sedimentation rates, high rates of recovery in soft intervals, and excellect microfossil preservation. If many sites contain such evidence for reversed polarity magnetizations in a limited paleontological range, this may be taken as a consistency test for the actual existence of corresponding reversed polarity intervals.

Reversed polarity magnetizations have been inferred in Albian sediments from two principal DSDP legs: leg 27 in the Indian Ocean and leg 48 in the Bay of Biscay. Below, these studies are reviewed in light of our new observations from the Albian Contessa section.

DSDP Leg 27 (Sites 260 and 263), Indian Ocean

At DSDP site 260 in the Gascoyne Abyssal Plain, reversed polarity magnetizations based on inclination-only data were reported in nannoradiolarian ooze of core 260-17-1 (121-140 cm; 311.21-311.40 m below seafloor (bsf)) even though only minimum levels of alternating field demagnetization (5 mT) were applied. The possibility that these reversed magnetizations are an artifact due to accidentally inverted core pieces can not be excluded from the available data. Dating this possible reversed polarity zone is difficult due to the poor preservation and patchy distribution of Cretaceous fossils. Calcareous nannofossils are both selectively dissolved and reworked. Both processes should produce a bias toward the assignment of nannofossil zones which are too old for Cretaceous age sediments, and this best explains the discrepancy of reported ages [*Bukry*, 1974; *Proto Decima*, 1974].

Bukry [1974] suggests a pre-Aptian age for the section below core 260-12 based on presence-absence criteria in the low-diversity assemblage. *Proto Decima* [1974], however, placed the section in the *Prediscosphaera cretacea* Zone (CC8) of middle Albian age based on a single occurrence of *Prediscosphaera cretacea* in core 260-12. In addition, assemblages from the deeper cores are similar to both those of core 260-12 and middle Albian assemblages at DSDP site 259 in the Perth Abyssal Plain.

Benthic foraminifers from core 260-12 and adjacent cores suggest a late Aptian to early Albian age [*Scheibnerova*, 1974]. The foraminiferal assemblage consists mostly of solution resistant calcareous and agglutinated species of broad stratigraphic range. Nevertheless, the benthic assemblages are clearly distinguishable from pre-Barremian assemblages and support the middle Albian age for core 260-17. This age interpretation places the possible site 260 core 260-17 reversed polarity zone within the stratigraphic range of the potential reversed polarity sequence at the Contessa Quarry. Using the above age interpretations, the sedimentation rate during the Albian at site 260 is at least 2-3 times greater than that recorded at the Contessa Quarry.

Similar intervals of reversed polarity magnetization (263-1: core 263-20-3, 26 cm to 263-3, 32 cm, 483.26-483.32 mbsf; 263-2: core 263-21-4, 6 cm to 263-21-4 11 cm, 519.56-519.61 mbsf; 263-3: core 263-23-2, 103 cm to 263-23-2 113 cm; 596.53-596.63 mbsf, 263-4: core 263-29-2, 99-101 cm; 765.99-766.01 mbsf) were found from claystones recovered at DSDP site 263 in the Cuvier Abyssal Plain [*Green and Brecher*, 1974; *Jarrard*, 1974].

The three uppermost intervals in cores 263-20 to 263-23 (263-1R, 263-1R, 263-3R) are dated as late Albian based on the occurrence of the nannoplankton *Eiffellithus turriseiffeli* [*Proto Decima*, 1974]. Macrofossils in cores 263-26 and 263-18 support the late Albian age [*Stevens*, 1974]. The presence of these possible reversed polarity zones led *van Hinte* [1976] to define a mid-Albian mixed polarity interval. As at site 260, the implied sedimentation rate for site 263 is at least 2-3 times greater than that from the Italian sections (Figure 24).

Another interval of reversed polarity magnetizations was also noted in core 29-2 (99-101 cm) and may be mid-Albian in age based on the presence of the nannofossil *Vagalapilla matalosa* in the overlying core [*Proto Decima*, 1974]. If correct, this event would correlate with the Contessa reversal sequence.

DSDP Leg 48 (Sites 400 and 402), Bay of Biscay

Paleomagnetic analyses of azimuthally unoriented cores recovered by DSDP leg 48 in the Bay of Biscay noted several intervals of reversed polarity magnetization [*Hailwood*, 1979]. At site 400, five short zones were defined in Aptian-Albian sediments (400-1R: core 400-62-3, 22-24 cm to 400-62-3, 105-107 cm, 657.23-658.06 mbsf; 400-2R: core 400-62-CC, 663.50 mbsf; 400-3R: core 400-63-2, 107-109 cm, 666.08 mbsf; 400-4R: core 400-63-4, 120-122 cm, 669.21 mbsf; 400-5R: core 400-72-4, 46-48 cm, 753.97 mbsf), although four of these "zones" are defined by only a single specimen each

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(400-2R, 400-3R, 400-4R, and 400-5R). Although each of these intervals was taken as indicating reversed geomagnetic field polarity [*Hailwood et al.*, 1979, 1980], a well-defined present-day overprint is absent and the possibility of inverted core pieces can not be dismissed. *Hailwood et al.* [1979], however, noted the coincidence of the Albian intervals with those of site 263. The two lower Albian intervals (400-3R and 400-4R) correlate with the Contessa intervals. The two upper intervals (400-1R and 400-2R) are contained within the *Eiffellithus turriseiffeli* Zone and do not correlate with those at Contessa.

Although rare, planktic foraminfers yield the best age for the lowermost interval at site 400 (400-5). This interval was placed within the *Globigerinelloides ferreolensis* Zone [*Hailwood*, 1979] by the recognition of three specimens of the nominate species in the core-catcher sample from core 400-72 which contains interval 440-5 [*Dupeuble*, 1979]. The core catcher sample of the overlying core (sample 400-71-CC), however, contained rare specimens of both *Globigerinelloides algerianus* and *Hedbergella trocoidea* [*Dupeuble*, 1979] indicating that, if present, the *G. algerianus* Zone lies somewhere within core 400-72. Therefore, interval 400-5 may correlate with the ISEA reversed polarity interval (upper Aptian) which falls within the *G. algerianus* Zone [*Tarduno*, 1990].

Two zones of reversed polarity were defined through paleomagnetic analyses of Aptian-Albian sediments at site 402 [*Hailwood*, 1979]. These zones (402-1R: 413.29-413.74 mbsf; 402-2R: 441.91-442.47 mbsf) have been confirmed and are discussed by *Tarduno* [1990]. Since a well-defined present-day overprint was not defined during demagnetization, the possibility of accidentally inverted core sections can not be excluded. The lowermost zone (402-2R), however, may correlate with the ISEA reversed polarity interval [*Tarduno*, 1990]. The upper interval (402-1R), falls within the *Ticinella bejaouansis* Zone [*Dupeuble*, 1979] which ranges from late Aptian to early Albian in age.

Summary

Taking the published paleomagnetic interpretations at face value, the "reversed" polarity zones used to define the "site 263" mixed polarity interval do not correlate with the potential reversed polarity zones at Contessa. The high implied sedimentation rates for site 263 are consistent with the paleoenvironmental interpretation calling for the influences of a large river [*Veevers et al.*, 1974]. The high sedimentation rates and the small stratigraphic thicknesses of the site 263 "reversed" polarity zones implies that these zones are artifacts, records of higher-frequency geomagnetic fluctuations than those potentially recorded at Contessa, or that sediment accumulation was not uniform.

The single reversed polarity zone at site 260 could be correlated with the Contessa sequence. If real the site 260 zone would best correlate with one of the thinner zones at Contessa. The two lower reversed polarity zones interpreted at site 400 can also be correlated with those at Contessa. If due to inverted cores, it is a curious coincidence that such mistakes should occur so close to the Contessa reversal sequence and not farther down the core (Figure 24).

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REFERENCES

Alvarez, W., and W. Lowrie, Upper Cretaceous paleomagnetic stratigraphy at Moria (Umbria Apennines, Italy): Verification of the Gubbio section, *Geophys. J. R. Astron. Soc.*, *55*, 1–17, 1978.

Alvarez, W., M.A. Arthur, A.G. Fischer, W. Lowrie, G. Napoleone, I. Premoli Silva, and W. Roggenthen, Late Cretaceous-Paleocene geomagnetic reversal time scale, V, Type section in the pelagic limestone sequence at Gubbio, Italy, *Geol. Soc. Am. Bull.*, *88*, 383–389, 1977.

Alvarez, W., T. Engelder, and P.A. Geiser, Classification of solution cleavage in pelagic limestones, *Geology*, *6*, 263–266, 1978.

Arthur, M.A., and A.G. Fischer, Late Cretaceous-Paleocene geomagnetic reversal time scale, I, Lithostratigraphy and sedimentology, *Geol. Soc. Am. Bull.*, *88*, 367–389, 1977.

Arthur, M.A., W.E. Dean, D. Bottjer, and P.A. Scholle, Rhythmic bedding in Mesozoic-Cenozoic pelagic carbonate sequences: The primary and diagenetic origin of Milankovitch-like cycles, in *Milankovitch and Climate, Part 1*, edited by A.L. Berger et al., pp. 191–222, D. Reidel, Norwell, Mass., 1984.

Bortolotti, V., P. Passerini, M. Sagri, and G. Sestini, The miogeosynclinal sequences, *Sediment. Geol.*, *4*, 341–444, 1970.

Bralower, T.J., and W.G. Siesser, Cretaceous calcareous nannofossil biostratigraphy of ODP leg 122 sites 761, 762 and 763, Exmouth and Wombat plateaus, N.W. Australia, *Ocean Drill. Program Sci. Results*, *122B*, in press, 1991.

Bukry, D., Coccolith stratigraphy, offshore western Australia, Deep Sea Drilling Project, leg 27, *Initial Rep. Deep Sea Drill. Proj.*, *27*, 623–630, 1974.

Channell, J.E.T., W. Lowrie, F. Medizza, and W. Alvarez, Paleomagnetic evidence for oroclinal bending in Umbria, Italy, *Earth Planet. Sci. Lett.*, *39*, 199–210, 1978.

Channell, J.E.T., W. Lowrie, and F. Medizza, Middle and Early Cretaceous magnetic stratigraphy from the Cismon section, northern Italy, *Earth Planet. Sci. Lett.*, *85*, 153–166, 1979*a*.

Channell, J.E.T., B. D'Argenio, and F. Horvath, Adria, the African promontory in the Mesozoic Mediterranean palaeogeography, *Earth Sci. Rev.*, *15*, 213–292, 1979*b*.

Channell, J.E.T., R. Freeman, F. Heller, and W. Lowrie, Timing of diagenetic haematite growth in red pelagic limestones from Gubbio (Italy), *Earth Planet. Sci. Lett.*, *58*, 189–201, 1982.

Coccioni, R., O. Nesci, M. Tramontana, F.C. Wezel, and E. Moretti, Descrizione di un Livello-Guida "Radiolaritico-Bituminoso-Ittiolitico" alla base delle Marne Fucoidi Nell'Appennnino Umbro-Marchigiano, *Boll. Soc. Geol. It.*, *106*, 183–192, 1987.

Coccioni, R., R. Franchi, O. Nesci, F.C. Wezel, F. Battistini, and P. Pallecchi, Stratigraphy and mineralogy of the Selli Level (Early Aptian) at the base of the Marne a Fucoidi in the Umbro-Marchean Apennines (Italy), in *Cretaceous of the Western Tethys, Proceedings 3rd International Cretaceous Symposium, Tübingen 1987*, edited by J. Wiedmann, pp. 563–584, Schweizerbart'sche, Stuttgart, Germany, 1989.

Coccioni, R., R. Franchi, O. Nesci, N. Perilli, F.C. Wezel, and F. Battistini, Stratigrafia, micropaleontologia e mineralogia delle Marne a Fucoidi delle sezioni di Poggio le Guaine e del Fiume Bosso (Appennino umbro-marchigiano), in *Atti 2° Convegno Internazionale "Fossili, Evoluzione, Ambiente" Pergola, 25-30 ottobre*, pp. 163-201, 1990.

de Boer, P.L., and A.A.H. Wonders, Astronomically induced rhythmic bedding in Cretaceous pelagic sediments near Moria (Italy), in *Milankovitch and Climate, Part 1*, edited by A.L. Berger et al., pp. 177-190, D. Reidel, Norwell, Mass., 1984.

Deutsch, E.R., and J.N. Prasad, Ordovician paleomagnetic results from the St. George and Table Head carbonates of western Newfoundland, *Can. J. Earth Sci.*, *24*, 1785-1796, 1987.

Dunn, W.J., and R.D. Elmore, Paleomagnetic and petrographic investigation of the Taum Sauk Limestone, Southeast Missouri, *J. Geophys. Res.*, *90*, 11,469–11,483, 1985.

Dupeuble, P.A., Mesozoic foraminifers and microfacies from holes 400A, 401 and 402A of the DSDP leg 48, *Initial Rep. Deep Sea Drill. Proj.*, *48*, 451–473, 1979.

21562202b, 1992, B1, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/91JB02257 by Curtin University Library, Wiley Online Library on [24/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Elmore, R.D., W. Dunn, and C. Peck, Absolute dating of dedolomitization by means of paleomagnetic techniques, *Geology, 13*, 558–561, 1985.

Erba, E., Aptian-Albian calcareous nannofossil biostratigraphy of the Scisti a Fucoidi cored at Piobbico (Central Italy), *Riv. Ital. Paleontol. Stratigr., 94*, 249-284, 1988.

Fischer, A.G., and W. Schwarzacher, Cretaceous bedding rhythms under orbital control?, in *Milankovitch and Climate, Part 1*, edited by A.L. Berger et al., pp. 163–175, D. Reidel, Norwell, Mass., 1984.

Fisher, N.I., T. Lewis, and B.J.J. Embleton, *Statistical Analysis of Spherical Data*, 329, Cambridge University Press, New York. 1987

Fisher, R.A., Dispersion on a sphere, *Proc. R. Soc. London, 217*, 295-305, 1953.

Gordon, R.G., A. Cox, and S. O'Hare, Paleomagnetic Euler poles and the apparent polar wander and absolute motion of North America since the Carboniferous, *Tectonics, 3*, 499–537, 1984.

Green, K.A., and A. Brecher, Preliminary paleomagnetic results for sediments from site 263, leg 27, *Initial Rep. Deep Sea Drill. Proj., 27*, 405-413, 1974.

Hailwood, E.A, Paleomagnetism of late Mesozoic to Holocene sediments from the Bay of Biscay and Rockall Plateau, drilled on IPOD leg 48, *Initial Rep. Deep Sea Drill. Proj., 48*, 305–339, 1979.

Hailwood, E.A., W. Bock, L. Costa, P.A. Dupeuble, C. Müller, and D. Schnitker, Chronology and biostratigraphy of northeast Atlantic sediments, DSDP leg 48, *Initial Rep. Deep Sea Drill. Proj., 48*, 1119–1141, 1979.

Hailwood, E.A., N. Hamilton, and G.E. Morgan, Magnetic polarity dating of tectonic events at passive continental margins, *Philos. Trans. R. Soc. London, 294*, 189-208, 1980.

Hambach, U., and K. Krumsiek, A magnetostratigraphic subdivision of the Upper Albian and Lower Cenomanian on the basis of drill cores from wells in the Ruhr area, *Geol. Jb., A113*, 401-425, 1989.

Harland, W.B., A.V. Cox, P.G. Llewellyn, C. Pickton, A.G. Smith, and R. Walters, R., *A Geologic Time Scale*, 131, Cambridge University Press, New York. 1982

Harland, W.B., R.L. Armstrong, A.V Cox, L.E. Craig, A.G. Smith, and D.G. Smith, *A Geologic Time Scale 1989*, 263, Cambridge University Press, New York. 1990

Heller, F., Rock magnetic studies of Upper Jurassic limestones from southern Germany, *J. Geophys., 44*, 525–543, 1978.

Herbert, T.D., R.F. Stallard, and A.G. Fischer, Anoxic events, productivity rhythms and the orbital signature in a mid-Cretaceous deep sea sequence from central Italy, *Paleoceanography, 1*, 495–506, 1986.

Jackson, M., Diagenetic sources of stable remanence in remagnetized Paleozoic cratonal carbonates: A rock magnetic study, *J. Geophys. Res., 95*, 2753–2761, 1990.

Jarrard, R.D., Paleomagnetism of some Leg 27 sediment cores, *Initial Rep. Deep Sea Drill. Proj., 27*, 415-423, 1974.

Kent, D.V., and F.M. Gradstein, A Cretaceous and Jurassic geochronology, *Geol. Soc. Am. Bull., 96*, 1419-1427, 1985.

Ladd, J.W., South Atlantic sea-floor spreading and Caribbean tectonics, Ph. D. thesis, Columbia Univ., New York, 1974.

Larson, R.L., and T.W.C. Hilde, A revised time scale of magnetic reversals for the Early Cretaceous and Late Jurassic, *J. Geophys. Res., 80*, 2586-2594, 1975.

Larson, R.L., and W.C. Pitman III, World-wide correlation of Mesozoic magnetic anomalies, and its implications, *Geol. Soc. Am. Bull., 83*, 3645–3662, 1972.

Leckie, R.M., Mid-Cretaceous planktonic foraminiferal biostratigraphy off central Morocco, Deep Sea Drilling Project leg 79, sites 545 and 547, *Initial Rep. Deep Sea Drill. Proj., 79*, 579-620, 1984.

Lowrie, W., Identification of ferromagnetic minerals in a rock by coercivity and unblocking temperature properties, *Geophys. Res. Lett., 17*, 159–162, 1990.

Lowrie, W., and W. Alvarez, Paleomagnetic evidence for rotation of the Italian Peninsula, *J. Geophys. Res., 80*, 1579-1592, 1975.

Lowrie, W., and W. Alvarez, Upper Cretaceous-Paleocene magnetic stratigraphy at Gubbio, Italy, III, Upper Cretaceous magnetic stratigraphy, *Geol. Soc. Am. Bull., 88*, 374–377, 1977.

Lowrie, W., and F. Heller, Magnetic properties of marine limestones, *Rev. Geophys., 20*, 171-192, 1982.

Lowrie, W., W. Alvarez, I. Premoli Silva, and S. Monechi, Lower Cretaceous magnetic stratigraphy in Umbrian pelagic carbonate rocks, *Geophys. J. R. Astron. Soc., 60*, 263–281, 1980*a*.

Lowrie, W., J.E.T. Channell, and W. Alvarez, A review of magnetic investigations in Cretaceous pelagic carbonate rocks, *J. Geophys. Res., 85*, 3597–3605, 1980*b*.

Lowrie, W., W. Alvarez, G. Napoleone, K. Perch-Nielson, I. Premoli Silva, and M. Toumarkine, Paleogene magnetic stratigraphy in Umbrian pelagic carbonate rocks: The Contessa sections, Gubbio, *Geol. Soc. Am. Bull., 93*, 414–432, 1982.

McCabe, C., and R.D. Elmore, The occurrence and origin of Late Paleozoic remagnetization in the sedimentary rocks of North America, *Rev. Geophys., 27*, 471-494, 1989.

Müller, C., Calcareous nannofossils from the North Atlantic (leg 48), *Initial Rep. Deep Sea Drill. Proj., 48*, 589–639, 1979.

Peck, C., R.D. Elmore, and R.L. DuBois, Early and late Paleozoic remagnetization of the Cambrian Peerless Formation in Colorado, *Phys. Earth Planet. Inter., 43*, 274-282, 1986.

Pratt, L.M., and J.D. King, Variable marine productivity and high eolian input recorded by rhythmic black shales in mid-Cretaceous pelagic deposits from central Italy, *Paleoceanography, 1*, 507-522, 1986.

Premoli Silva, I., G. Napoleone, A.G. Fischer, L.M. Pratt, E. Erba, and B. Quadrio, A mid-Cretaceous event as cored in the "Scisti a Fucoidi" at Piobbico, Italy, *Boll. Soc. Paleontol. It., 22*, 187, 1983.

Proto Decima, F., Leg 27 calcareous nannoplankton, *Initial Rep. Deep Sea Drill. Proj., 27*, 589–621, 1974.

Scheibnerova, V., Aptian-Albian benthonic foraminifera from DSDP leg 27, sites 259, 260 and 263, eastern Indian Ocean, *Initial Rep. Deep Sea Drill. Proj., 27*, 697–741, 1974.

Schwarzacher, W., and A.G. Fischer, Limestone-shale bedding and perturbations of the Earth's orbit, in *Cyclic and Event Stratification*, edited by G. Einsele and A. Seilacher, pp. 72–95, Springer-Verlag, New York, 1982.

Sissingh, W., Biostratigraphy of Cretaceous calcareous nannoplankton, *Geol. Mijnbouw, 56*, 37–65, 1977.

Sliter, W.V., Aptian anoxia in the Pacific basin, *Geology, 17*, 909-912, 1989.

Sliter, W.V., Cretaceous planktonic foraminiferal biostratigraphy and paleoceanographic events in the Pacific Ocean with emphasis on indurated sediments, in *Centenary of Japanese Micropaleontology*, edited by T. Saito and K. Ishizaki, pp. Terra Scientific, Tokyo, in press, 1991.

Stevens, G.R., Leg 27 Cephalopoda, *Initial Rep. Deep Sea Drill. Proj., 27*, 983-986, 1974.

Tarduno, J.A., Albian reversed polarity intervals within the Cretaceous Normal Polarity Superchron?, *Eos Trans. AGU, 70*, 1073, 1989.

Tarduno, J.A., Brief reversed polarity interval during the Cretaceous Normal Polarity Superchron, *Geology, 18*, 683-686, 1990.

Tarduno, J.A., W.V. Sliter, T.J. Bralower, M. McWilliams, I. Premoli Silva, and J. Ogg, M-sequence reversals recorded in DSDP sediment cores from the western Mid-Pacific Mountains and Magellan Rise, *Geol. Soc. Am. Bull., 101*, 1306–1319, 1989.

Thierstein, H.R., Lower Cretaceous calcareous nannoplankton biostratigraphy, *Abh. Geol. Bundesanst. Austria, 29*, 1–52, 1973.

Tornaghi, M.E., I. Premoli Silva, and M. Ripepe, Lithostratigraphy and planktonic foraminiferal biostratigraphy of the Aptian-Albian "Scisti a Fucoidi" in the Piobbico core, Marche, Italy: Background for cyclostratigraphy, *Riv. Ital. Paleontol. Stratigr., 95*, 223–264, 1989.

Urrutia-Fucugauchi, J., Paleomagnetic study of the Cretaceous Morelos Formation, Guerrero State, southern Mexico, *Tectonophysics, 147*, 121–125, 1988.

VandenBerg, J., and A.A.H. Wonders, Paleomagnetism of Late Mesozoic pelagic limestones from the southern Alps, *J. Geophys. Res., 85*, 3623–3627, 1980.

VandenBerg, J., C.T. Klootwijk, and A.A.H. Wonders, The Late Mesozoic and Cenozoic movements of the Umbrian Peninsula: Further paleomagnetic data from the Umbrian sequence, *Geol. Soc. Am. Bull., 89*, 133–150, 1978.

van Hinte, J.E., A Cretaceous time scale, *AAPG Bull., 60*, 498–516, 1976.

Veevers, J.J. *et al.*, Initial Reports of the Deep Sea Drilling Project, *27, U.S. Government Printing Office, Washington, D.C.*, 1974.

Wiegand, G.E., Cretaceous nannofossils from the northwest African margin, DSDP leg 79, *Initial Rep. Deep Sea Drill. Proj., 79*, 563–578, 1984.

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